



CATHOLIC UNIVERSITY OF GHANA, FIAPRE

FACULTY OF COMPUTING, ENGINEERING AND MATHEMATICAL SCIENCES

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DEPARTMENT OF COMPUTING AND INFORMATION SCIENCES

**WELLTRACK: A WEB-BASED WELLNESS MANAGEMENT SYSTEM FOR POLICE
OFFICERS**

BY

INUSAH ADAMA

(UGW0202210054)

SEDZRO REBECCA

(UGW0202210055)

**A PROJECT SUBMITTED TO THE FACULTY OF COMPUTING, ENGINEERING AND
MATHEMATICAL SCIENCES, IN PARTIAL FULFILMENT OF THE REQUIREMENT
FOR THE AWARD OF BACHELOR OF SCIENCE DEGREE IN COMPUTER SCIENCE**

DECLARATION

Students' Declaration

We declare that this is the result of our research and we have duly given credit where appropriate, and the references are made accordingly.

.....

INUSAH ADAMA

(ID No.: UGW0202210054)

.....

Date

.....

SEDZRO REBECCA

(ID No.: UGW0202210055)

.....

Date

Supervisors' Declaration

I hereby declare that I have checked this project and, in my opinion, this project is adequate in terms of scope and quality. I confirm that the student has my permission to present for assessment.

.....

Mr. Steven Okyere-Gyamfi

.....

Date

.....

Mr. Amankona Vincent

.....

Date

Head of Department's Declaration

I hereby declare that the preparation of this research project was in accordance with the supervision guidelines of research projects laid down by the Catholic University Ghana, Fiapre.

.....

Dr Afful Ekow Kelly

.....

Date

Dean's Declaration

I hereby declare that the preparation of this research project was in accordance with the guidelines and supervision of research projects laid down by the Catholic University College of Ghana, Fiapre.

.....

Dr. Alex K. Peprah

.....

Date

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DEDICATION

This project is wholeheartedly dedicated to Almighty God, whose grace, wisdom, and strength have guided us through every stage of this academic journey. We also dedicate this work to our beloved families, especially our parents, for their unwavering support, encouragement, and sacrifices that have enabled us to reach this far.

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ABSTRACT

A wellness management system is a web-based software platform designed to monitor, manage, and mitigate occupational stress among specialized personnel. Law enforcement officers face significant challenges in maintaining mental health due to exposure to trauma, long working hours, and the stigma associated with seeking help. This project seeks to design and develop a responsive web application, **WellTrack**, that provides a centralized platform for stress assessments, confidential counseling modules, peer support forums, and wellness resource dissemination. The development of such a system is critical for modernizing police welfare infrastructure, which currently relies on fragmented or paper-based reactive measures. Existing wellness programs often lack the confidentiality and real-time accessibility required to encourage officer participation features which our proposed system addresses. The methodology and tools used for the development and deployment of this system follow the **AGILE METHODOLOGY** for its iterative and user-centered development process. Tools and technologies include Node.js (Express.js), MongoDB, TailwindCSS, Next.js and JavaScript. System testing showed the platform successfully facilitated stress tracking, allowed administrators to analyze department-wide wellness trends, and provided secure communication channels for peer support. The system successfully implements a proactive digital approach to mental health, fostering a more resilient and supportive work culture within the police service.

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ABBREVIATIONS

WMS – Wellness Management System

UI – User Interface

AI – Artificial Intelligence

API – Application Programming Interface

GPS – Ghana Police Service

PTSD – Post-Traumatic Stress Disorder

PHQ-9 – Patient Health Questionnaire (Stress Assessment Tool)

ERD – Entity Relationship Diagram

OOADM – Object-Oriented Analysis and Design Methodology

UML – Unified Modeling Language

SQL – Structured Query Language

DB – Database

CSS – Cascading Style Sheets

PHP – Hypertext Preprocessor

SDLC – Software Development Life Cycle

CHAPTER ONE

INTRODUCTION

Before technology came into existence, dealing with the mental and emotional well-being of police officers was very difficult. Law enforcement agencies didn't have the tools to quickly detect or track the rising levels of occupational stress or trauma among personnel; this means that by the time they realized an officer was suffering from burnout or mental health crises, it had often already affected their performance and personal lives. Communication between units and medical welfare departments was limited; this means responses were uncoordinated and ineffective.

In recent years, technology has drastically enhanced the way wellness is monitored and how mental health awareness is created, with the use of real-time data collection and accurate analysis that process massive amounts of wellness metrics. By analyzing this data, this software can detect patterns and anomalies that might signal the start of severe psychological distress. This early detection capability gives welfare authorities a crucial head start, allowing them to take action before an officer's mental health deteriorates widely[1].

In conclusion, this project, WellTrack: A Wellness Management System, is a web-based system designed to detect, track, and respond to the emergence and spread of occupational stress and mental health challenges at a departmental or regional level, and to predict future possibilities of officer burnout. The system integrates technologies, scientific methodologies, and monitoring of potential wellness crises to minimize their impact on officer health, police operations, and the community[2].

1.1 Background Of Study

The term "wellness" in professional contexts has evolved to describe a continuous state of physical, mental, and social well-being, particularly in high-stress occupations. Historically, the concepts of "stress" and "burnout" were utilized alternatively in many social and medical contexts during the 17th and 18th centuries. But however, these words have developed throughout time, and new ideas have emerged. The terms occupational stress, secondary trauma, and post-traumatic stress disorder (PTSD) express how frequent and severe mental health challenges are now compared to previously[3]. To differentiate between these terms, occupational stress occurs when job demands exceed an officer's capabilities. Secondary trauma is an unanticipated increase in distress from witnessing traumatic events. Burnout is a state of exhaustion that spreads across an officer's career lifespan. While a mental health crisis is an acute episode that expands to impact both professional and personal life.

Mental health challenges have been a threat factor to law enforcement history, starting with early policing eras where high mortality and physical danger were the primary focus. However, the absence of systematic data collection and reports made it challenging to understand the psychological spread of trauma and effective containment strategies. The responses were primarily based on isolation and "toughing it out" practices, highlighting the need for a structured approach to monitoring officer wellness[2].

Few years ago, global events and rising crime rates highlighted the intense pressure on the Ghana Police Service. Occupational stress is a highly impactful psychological condition caused by exposure to trauma and long hours. The stress rapidly spreads across the force, leading to what many experts declare a "silent pandemic" of mental health struggles. It is transmitted primarily

through high-pressure environments and has a wide range of symptoms, from mild anxiety and irritability to severe depression, particularly in officers in high-risk units. Its high prevalence and potential for severe outcomes overwhelmed traditional welfare systems, leading to reduced productivity and widespread social disruption within the force. In response to this, global health organizations have coordinated efforts to improve mental health screening and implement preventive measures like counseling. This situation has highlighted the importance of robust wellness infrastructure, real-time data monitoring, and institutional cooperation in managing officer health. Despite significant advancements in psychology, stress continues to pose challenges, underscoring the need for ongoing vigilance and adaptation in public health strategies. "The lack of proactive wellness systems has highlighted critical weaknesses in police welfare, such as the lack of effective surveillance and rapid response mechanisms to psychological threats,"[4].

The study of wellness management systems, viewed through the lens of historical law enforcement challenges, highlights the necessity of continuous evolution and adaptation in welfare responses. From ancient practices to the sophisticated digital systems of today, it shows the importance of systematic monitoring, data collection, and predicting future wellness trends.

1.2 Problem Statement

Occupational stress is a recurring threat factor to law enforcement for centuries, leading to high burnout rates, economic disruption through loss of man-hours, and societal disturbance. Various studies on police welfare have revealed that traditional systems are not always prepared to handle large-scale psychological distress, particularly the inadequacy of real-time monitoring and effective response mechanisms. The use of traditional manual reporting methods and face-to-face counseling proved ineffective to track the pattern of these mental health issues in real time, leading

to delayed responses and worsening the internal crisis. The wellness management system problem statement starts from the failures and shortcomings experienced in past welfare programs, mainly in their detection, response, coordination, and data-sharing mechanisms.[5]

This situation shows the essential need for a robust, technology-driven wellness management system that can integrate real-time data from diverse sources, including individual assessments, peer support forums, and professional counselors. A wellness management system would provide early warning signals, track stress trends, and offer actionable insights to mitigate future mental health crises. The primary challenge lies in developing a scalable system that not only responds to ongoing distress but also anticipates future threats, ensuring rapid response and resource allocation during a welfare crisis. With the presence of a real-time system that can enable an accurate display or visualization of this data, officer wellness can be monitored better and closer wherever and whenever it is.

1.3 Research Questions

- a) How will this wellness management system effectively track, report, and mitigate the spread of occupational stress?
- b) What type of information should be prioritized to enable prompt wellness response and decision-making?
- c) What methods would be most effective for collecting real-time wellness data from a wide range of officers?
- d) How will this system ensure easy access and enhance user engagement while maintaining anonymity?

- e) How responsive should the system handle notifications and alerts to inform users when high-risk stress levels are detected?

1.4 Aims and Objectives

Project Objective:

To develop an effective wellness management system that enhances real-time stress surveillance and coordinated response mechanisms to effectively identify, track, and mitigate the spread of mental health challenges among officers.

General Objective:

To strengthen the institutional health awareness and response capabilities by implementing a wellness management system that improves early detection of psychological distress.

Specific Objectives:

- a) Real-Time Data Integration: To develop a centralized platform that integrates data from various assessments for real-time monitoring of wellness trends.
- b) Public Awareness and Engagement: To implement communication strategies that inform officers about mental health resources and wellness forecasts.

1.5 Project Scope

This study covers areas such as the design phase and development phase of the wellness system that enables real-time monitoring. The system seeks to support officers, welfare departments, and police organizations by providing real-time information to aid in decision-making and accurate wellness response. The functionality of the system is as follows:

The functionality of the system is as follows:

- a) **Data Collection & Integration:** Gathering and integrating data from reliable assessment tools and officer inputs.
- b) **Real-Time Data Monitoring:** Enables real-time tracking of stress metrics with frequent updates on a dashboard to provide an overview of the ongoing wellness situation.
- c) **Data Analyzing:** Allows generation of detailed reports on wellness data that can be used for analysis, decision-making, and planning by the Welfare Department.
- d) **Organized to support various user access levels (Officers, Counselors, Admin) and the ability to send notification updates**
- e) **Data Security & Privacy:** strict data protection protocols to ensure adherence to laws governing the privacy of health data and user anonymity.
- f) **Scalability and Flexibility:** Management of both small-scale unit monitoring and large-scale regional or national police data.
- g) **User Management:** Create secure login and role-based access for different users.
- h) **Data Dashboard:** Present a dashboard with key metrics, trends, and analytics for monitoring health statistics.
- i) **Location-Based Data Visualization:** Use mapping tools to visualize stress density and risk zones within different police stations.

1.6 Project Justification & Motivation.

The existing manual and decentralized wellness surveillance systems are significantly inefficient, leading to substantial discontent among police welfare professionals and officers. The implementation of a digital real-time wellness management system has the potential to transform stress tracking, increase the effectiveness of psychological responses, and provide a more proactive welfare infrastructure. Traditional monitoring methods inherently pose the risk of delayed information processing, insufficient data gathering, and restricted predictive capabilities—critical shortcomings that can be comprehensively addressed through an integrated, technologically advanced platform. "Existing wellness mechanisms lack enforcement and rely heavily on institutional will and technical capacity. Building robust wellness systems requires coordinated governance, advanced technology, and sustainable support"[2].

As computer science students, we are committed to continuous learning and problem-solving, embracing the complex challenge of developing a sophisticated technological solution that can potentially save lives and mitigate the impact of mental health crises. Through this project, our ultimate aim is to leverage our technical skills to create an innovative solution that streamlines wellness monitoring and contributes meaningfully to public health technology while demonstrating the transformative potential of integrated digital systems[6].

The justification for this project is rooted in the continuously evolving landscape of occupational health management, emphasizing the critical role of real-time, data-driven approaches. Historically, officer monitoring has been characterized by significant delays and reactive rather than proactive strategies. The accessibility and immediacy of comprehensive wellness data have

consistently been challenges expressed by welfare officers and researchers worldwide. To effectively combat these challenges, the development of a robust real-time wellness management system becomes not just beneficial, but imperative. The implementation of this system will lead to immediate tracking of stress spread, comprehensive data integration, and enhanced decision-making tools for rapid intervention[7].

1.7 Conceptual Framework

The conceptual framework for the development and implementation of this real-time wellness management system aims to revolutionize police welfare by transitioning from fragmented, reactive approaches to an integrated, proactive digital platform. The goal of this project is to address the critical challenges associated with traditional monitoring methods, including delays in data collection and uncoordinated response mechanisms[7].

At its core, the framework highlights the significance of leveraging advanced technology to enhance the speed and accuracy of wellness surveillance[8]. Through a centralized platform, we will collect and analyze health data in a structured manner. By implementing analytics, patterns and potential crises become quickly identifiable, streamlining the process of decision-making for welfare providers and officials.

Furthermore, the framework places a high priority on robust security to safeguard sensitive data against unauthorized access. We will implement sophisticated user authentication protocols and role-based access (Officer, Counsellor, Admin) to ensure the confidentiality and integrity of stored information. This approach facilitates both privacy protection and efficient data sharing during critical situations[9].

Lastly, the framework includes provisions for continuous improvement. We will develop an analytics component to analyze system performance and response effectiveness, ensuring the system remains aligned with emerging health challenges and best practices in wellness surveillance[10].

WellTrack - Web-Based Wellness Management System



Figure 1.0: A conceptual design for a wellness management system

1.8 Project Organization

The remaining chapters of this study will be coordinated as follows:

- a) **Chapter 2** provides a comprehensive overview of real-time wellness management systems, encompassing various types and aspects related to mental health surveillance technologies.
- b) **Chapter 3** outlines the methodology employed, encompassing the hardware and software requirements, addressing the research questions, and designing the system architecture.
- c) **Chapter 4** presents the analysis and design of the proposed system, including an examination of its current state and security requirements.
- d) **Chapter 5** focuses on the implementation and testing of the new system, beginning with the implementation process and delving into the specifics of the main functions of the WellTrack system.

CHAPTER TWO

LITERATURE REVIEW

INTRODUCTION

Prior to the integrated wellness management systems, many police institutions have historically depended on conventional welfare surveillance techniques. Although these conventional methods were essential for monitoring and controlling officer health, they have a number of drawbacks that have been extensively reported in the reviewed literature.

Geographical Limitations is one of the major issues faced by traditional police welfare systems because this approach relied on centralized reporting techniques that required manual data submission from specific police stations and regional commands. In the review of this approach, geographical limitation makes it difficult to track stress patterns in remote or rural precincts. Limited digital health infrastructure in certain regions hindered real-time data collection, delaying the detection of mental health crises and intervention efforts[10]. For instance, during periods of high civil unrest, traditional welfare systems struggled to collect timely data from officers in the field, contributing to slower counseling measures and higher burnout rates.

Manual Welfare Reporting Systems is another challenge faced by the traditional police settings that have historically depended on a manual reporting approach where welfare officers report cases and documents of psychological distress manually. Despite being fundamental, this method has serious problems with timeliness and data accuracy, as it often leads to delayed detection of mental health trends and inconsistent data quality due to varying reporting standards across different departments. "Traditional welfare systems, relying on manual reporting from unit commanders,

have historically suffered from reporting delays of weeks to months, severely limiting their utility for rapid stress intervention"[10].

Laboratory and Clinical Facilities previously maintained paper-based systems for tracking officer medical results and psychological evaluations. These physical record-keeping methods posed substantial challenges in data sharing and analysis[11]. Researchers have documented that paper records significantly hindered rapid response capabilities during crises, as information sharing between medical officers and police authorities was time-consuming and prone to errors. Paper-based systems presented a major bottleneck in wellness surveillance, with one study showing that only 79% of critical mental health cases were successfully communicated to the appropriate welfare authorities within the recommended timeframe.

2.1 Internet and Web Applications: Wellness Management System

This segment focuses on the development of a web-based wellness management system designed to optimize surveillance and response of real-time occupational stress. The existing surveillance systems have been dependent on manual data collection and periodic reporting, which leads to delays in crisis detection and challenges in data integration[12]. This web-based wellness system aims to work on these limitations by providing a centralized platform for real-time data collection, analysis, and reporting. By utilizing structured assessments and predictive analytics, this system enhances the accuracy of stress tracking and facilitates future mental health assessment.

2.1.1 Introduction to this System "Web-based Application: Wellness Management System"

The paper discusses the development of a wellness management system that aims to transform traditional welfare methods into an integrated web-based platform accessible to officers, counselors, and other relevant stakeholders [12], [13].

The system is designed to address the limitations of manual reporting processes and fragmented welfare networks, providing a centralized platform for real-time wellness monitoring and distress detection across multiple police stations[11]. This digital transformation enables rapid data collection, analysis, prediction, and response coordination through automated processes and standardized protocols.

The project focuses on creating a web-based monitoring system that integrates various data sources, including self-assessments, peer support logs, and clinical reports, making it easier to detect and track psychological patterns. The traditional method highlights the critical need for a system that can handle sensitive health data while ensuring secure and efficient information sharing between different welfare units. Modern wellness surveillance requires robust digital infrastructure capable of processing diverse data types while maintaining high data security and accessibility.

2.1.2 Methodologies Used in The System "Web-based Application: Wellness Management System"

The structured Agile methodology used in the creation of the web-based wellness management system guarantees effective data processing and collecting for real-time surveillance of stress levels to improve crisis identification and response. The system uses the Agile Methodology to

facilitate continuous testing, iterative development, and flexibility in integrating officer feedback. Agile makes sure that the system changes in response to actual data from counselors, administrators, and the officers themselves[14]. The technology improves wellness monitoring speed and accuracy through the use of agile methodology, which helps welfare professionals make better decisions and respond faster.

2. 1. 3 Good Features Of "Web-based Application: Wellness Management System"

This feature enables real-time data collection from multiple assessment modules that ensures timely updates on officer stress levels.

A user-friendly dashboard provides visual representations of wellness trends through features that enhance data interpretation, making it easier to track the spread of burnout.

This system can analyze historical assessment data to predict potential mental health crises by helping in early intervention and resource allocation.

This system generates automated alerts for counselors and the administration when a significant stress peak is detected. These notifications help in preventive measures and support enforcement[15].

The system implements robust security protocols, including encryption and user authentication, to protect sensitive mental health data from unauthorized access and cyber threats.

2.1.4 Bad Features of Existing Wellness Systems

Insufficient explanation of the wellness system's particular features and capabilities. Lack of privacy and data security safeguards in place within the system. The wellness system's usability and acceptance may suffer if user interface design and accessibility concerns are not addressed[15]. Lack of information about performance and scalability considerations, which are essential for managing a high volume of personnel data. Ignoring problems with the system validation and testing procedure to guarantee the dependability and correctness of the psychological data.

2.1.5 “A web-based wellness management system for Police Officers”

A web-based wellness management system for law enforcement institutions and welfare organizations designed to enhance stress surveillance and mental health response through integrated digital technologies.

2.2 Introduction of “A web-based wellness management system for General Police Use”

The web-based pandemic monitoring system represents an innovative approach to global disease surveillance and public health management, which create a comprehensive method for detecting, tracking, and responding to disease outbreaks across multiple locations.

This system will be developed using opensource software and OOADM methodology[14].

System validation tests confirm effectiveness in managing and sharing health data.

2. 2. 1 The Methodologies Used In “A web-based wellness management system for General Police Use.”

The software engineering paradigm used to construct this wellness management system is called Object-Oriented Analysis and Design Methodology (OOADM). This methodological approach guarantees a solid system architecture and methodical development that supports intricate wellness surveillance requirements[16].

The need for a solution to the difficulties police welfare organizations encounter in identifying, monitoring, and reacting to officer distress is the foundation for the creation of the wellness system. Frameworks and tools, such as Next.js for web application development and MongoDB for database administration, are used in the system's implementation. The use of robust software in the creation of the web-based wellness management system is also highlighted[17]. The benefits of modern technologies, such as encouraging institutional cooperation and guaranteeing ongoing advancements through security updates, are what motivated this decision. The constructed system's viability is evaluated using system validation tests, which demonstrate the application of

a validation technique to appraise the system's efficacy in tackling issues pertaining to stress identification and wellness decision-making.

2.2.2 The Good Features of "A Web-Based Wellness Management System for General Police Use"

The paper presents the development of a web-based wellness management system designed to provide real-time tracking and reporting of stress levels, addressing the need for an effective, efficient, reliable, and easily accessible system for welfare management. The implementation of this system improves the traditional manual methods of data collection, ensuring timely and accurate information dissemination for counselors and policymakers. The system reduces delays in distress detection and response, thereby enhancing institutional preparedness and intervention strategies[18].

The system provides secure access to wellness-related data, increasing institutional impact and enhancing the ability of welfare units to analyze and respond to burnout effectively. It enables users to have 24/7 access to wellness resources, facilitating rapid information retrieval and making it convenient for officers to stay informed. The web-based wellness management system ensures the preservation and consolidation of health data, reducing data loss and maintaining historical records for future reference. This feature enhances long-term planning and research efforts in police psychology[14].

It allows multiple users, including counselors and administrators, to access real-time wellness data simultaneously.[19].

2.2.3 The Bad Features of “A Web-Based Wellness Management System for General Police Use”

Despite its many benefits, the web-based wellness management system has many drawbacks that could affect its overall usefulness. Among the challenges is data security; the system may be vulnerable to unauthorized access because it manages sensitive mental health data. This system requires a consistent internet connection to function properly, which could cause issues in remote stations with poor network infrastructure. Because this system relies on precise data, if officers don't provide honest assessments, it may be unable to track wellness trends accurately, resulting in delayed interventions.

2.2.4 A Web-Based Wellness Management System for West Africa (Ghana Police)

The development of this system addresses the need to improve officer welfare and stress identification; this paper describes the creation of a web-based wellness management system designed especially for local contexts. The system is intended to tackle the particular problems, such as the lack of a centralized monitoring platform and restricted healthcare infrastructure in local precincts[20].

The system incorporates real-time data collection from stations and welfare organizations to ensure timely reporting during mental health emergencies. Counselors may more easily monitor trends thanks to its interactive dashboard, which uses statistical graphs to illustrate stress levels.[19].

2.2.5 The methodologies used "A Web-Based Wellness Management System for West Africa."

The creation of a web-based wellness management system for Africa that tracks, analyzes, and visualizes occupational stress in real-time is described. Through a web interface, the system gives users an interactive platform to access wellness data[21].

Administrators may manage personnel data, track stress patterns, and create reports using the system's administrative interface. The system integrates data visualization features, such as graphs and statistical charts, to enhance data comprehension. A scalable database design is used in the system's backend using MongoDB. The web-based interface ensures a responsive user experience, following Agile development principles[22].

2.3 Good features of the system "Web-Based Wellness Management System for West Africa."

The system provides an administrative interface for managing wellness-related data, including modifying and deleting assessment reports. It can be used by administrators to monitor stress patterns, establish access control measures, and generate statistical reports. By categorizing wellness data by department and severity, it enables rapid data retrieval. The system supports automated backups, which ensure data integrity. It also has the ability to create interactive dashboards to enhance real-time decision-making.

2.3.1 Bad features of the "Web-Based Wellness Management System for West Africa."

This system requires adequate internet connectivity, which may limit accessibility in rural police posts. The system's ability to monitor a large force effectively may be diminished if users encounter delays in receiving data. It requires constant input from officers and welfare agencies. Inconsistent submissions may lead to inaccurate predictions. Due to its reliance on server availability, the system is susceptible to outages. The collection of sensitive health information may also cause privacy concerns among officers if not properly managed.

2.3.2 Developing A User-Friendly Wellness Monitoring System for Tracking Stress

The system highlights the challenges faced in existing welfare platforms, which often suffer from complexity and limited accessibility. The system highlights how crucial it is to create a wellness tracking tool that is easy to use for officers and counselors. This includes creating an intuitive web interface. The objective is to offer a simplified platform that facilitates effective wellness management[23].

2.3.3 Methodologies used in this system

The system methodology is interactive and responsive, and is implemented through modern web technologies to ensure a user-driven and customizable interface. The methodologies employed in developing the user-friendly **Wellness Monitoring System for Tracking Stress** include:

- a) **Iterative Development Process:** Incorporating feedback from officers and counselors to continuously refine features [22].

- b) **Usability Techniques:** Using user testing and heuristic evaluation to ensure interface simplicity for technical and non-technical users.
- c) **Data Integration:** Aggregating data from multiple assessment sources using a structured framework.
- d) **Classification of Wellness Services:** Categorizing services based on user needs, such as administrators requiring analytics and officers needing personal tracking[18].

2.3.4 Good Features

The system is designed for ease of use, providing a user-friendly interface for seamless interaction with wellness data. It adopts a service-based framework, enabling integration with multiple data sources. A key component is the administration module for data management and reporting. Through the visualization module, users can access stress charts and filter information based on department or time frame.

2.3.5 Bad features

The publication of this system may have some limitations or areas that could be enhanced, such as:

- a) **Lack of comparative analysis:** Existing systems may lack evaluation against defined advantages.
- b) **Restricted information on implementation Challenges:** Insufficient discussion of technical challenges encountered during development.
- c) **Poor comprehensive evaluation:** The system may not include a thorough assessment of its effectiveness in improving wellness tracking.

2.4.0 Summary and Discussion of This Chapter

The design and development of web-based wellness management systems have become essential in law enforcement today. The existing literature pinpoints key aspects such as system development, essential functionalities, and implementation challenges. The literature review underscores the role of digital systems in processing and visualizing wellness data. The evolution from basic manual welfare to advanced analytical systems shows the demand for data-driven wellness management.

Conclusion

In conclusion, the literature review identified the vital role of digital health technologies in modern wellness tracking. While progress has been achieved, continuous development is necessary to address data accuracy and privacy. A real-time visualization chart and analysis, powered by structured data, enhances the system's ability to detect stress peaks. After reviewing multiple related systems, this platform will adopt agile development methodologies, incremental implementation, and stringent security measures to create a robust wellness monitoring solution. [18], [19].

CHAPTER THREE

METHODOLOGY

3.0 The overview of the Proposed System

A method is a methodical approach to achieving a certain goal or resolving a specific issue. It consists of a series of actions and processes that have been tried and found to work. To monitor officer well-being more effectively and efficiently, for instance, the police welfare sector may use data collection, processing, and analysis.

Methodology therefore refers to the systematic framework guiding researchers in designing, conducting, and analyzing investigations. It encompasses processes, techniques, and tools to gather and interpret data ensuring the reliability and validity of research findings[12].

Moreover, Methods may be construed as the sequential actions undertaken to achieve a specific objective. It is imperative to recognize that each project or research work possesses its unique approach or methodology in attaining its objectives. The development of this web-based wellness management system (WellTrack) follows a structured and systematic approach to ensure that the system is efficient, user-friendly, scalable, and technically robust. This methodology section outlines the key phases and activities involved in designing, developing, and deploying the system, ensuring its effectiveness in real-time wellness tracking[10].

3.1 Project Method Adopted and Justification.

The Agile Scrum methodology is the most suitable for designing the WellTrack system due to its flexibility, incremental delivery, effective risk management, and focus on continuous improvement and quality assurance.

Agile Scrum methodology is a step-by-step approach to software development that is both iterative and incremental, focusing on flexibility and rapid delivery of functional software. The approach involves dividing the project into small parts, typically lasting two to four weeks, during which specific system features like stress assessments or peer forums are developed, tested, and The Agile Scrum methodology's iterative nature allows for continuous improvements and adjustment of the wellness management system based on constant updates and evolving psychological needs, making it perfect for sensitive and dynamic mental health environments. The flexibility of Agile ensures efficient adaptation to emerging welfare policy changes and technological advancements.

The incremental delivery of this approach allows the early and continuous release of functional components such as stress tracking modules, resource dashboards, and counseling request forms. This early delivery can enable timely issue resolution, better progress tracking, and early access to critical wellness functionalities, enhancing the system's reliability and effectiveness.

Risk management is inherently supported by the Scrum iterative approach, as each sprint focuses on delivering a potentially shippable product increment, enabling frequent risk assessment and mitigation regarding data privacy and system robustness. A continuous improvement is promoted through regular retrospectives, enabling the team to identify areas for enhancement in the user

interface or assessment algorithms. Quality assurance is integrated throughout the Scrum development lifecycle, ensuring each increment is thoroughly vetted before deployment.

Various Phases of The Agile Model are as follows:

Here is a graphical illustration of the Agile Model:

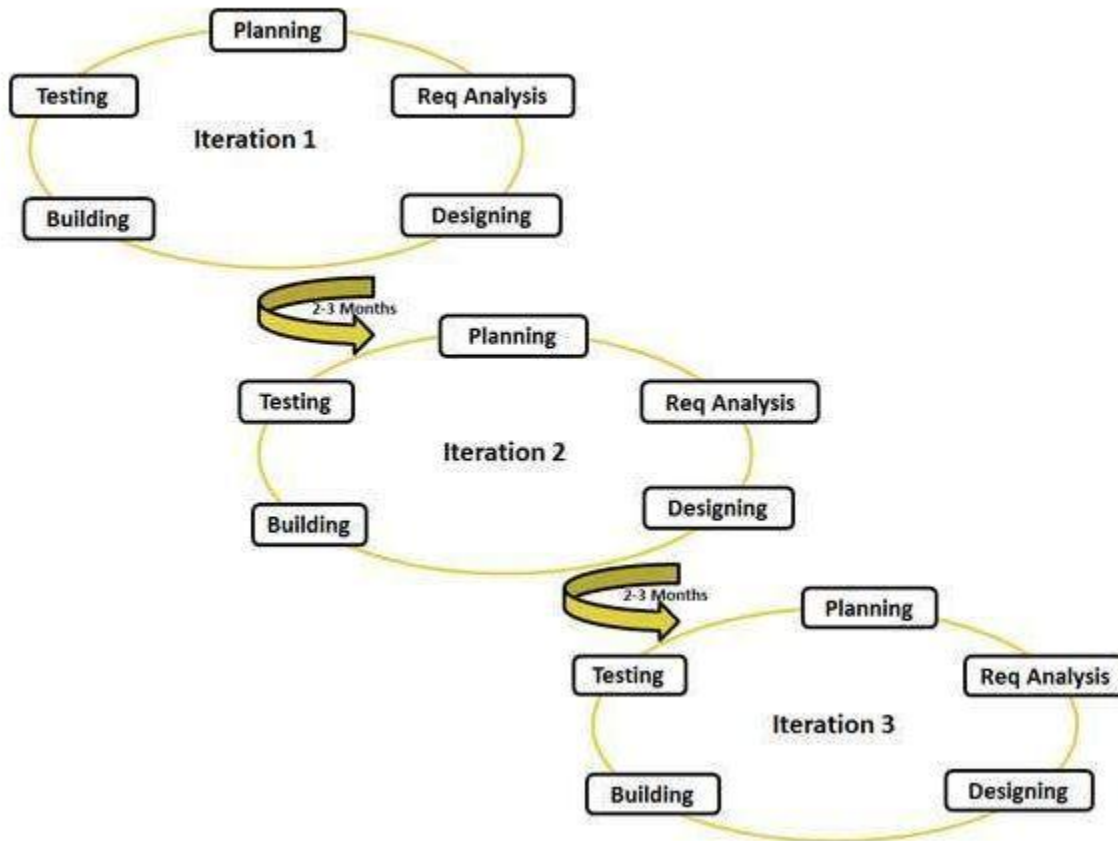


Figure 3.0: Various phase of the development

3.1.1 Various Phases of The Development

The following are the detailed explanations of the four phases of the development:

1. Requirements:

We concentrated on gathering the system's fundamental requirements during this phase, identifying the need for confidentiality and ease of use for officers. A system that hosts wellness monitoring functions centrally is developed to improve accessibility.

2. Design and Development

The architecture is established, and features for each increment (e.g., the assessment engine) are designed. This involve creating UML diagrams like ERDs, use case diagrams, and sequence diagrams to visualize the flow of mental health data. The UI design focuses on ensuring the system is intuitive and supportive.

3. Testing

Each increment is rigorously tested to ensure that features like the anonymous forum or assessment scoring function as expected and are free from defects.

4. Implementation

This involves deploying the system into a live environment for the Ghana Police Service. It includes user training and system documentation to ensure officers and counselors understand how to use the platform effectively. WellTrack is implemented using Node.js (Express.js), JavaScript, and MongoDB.

5. 3..1.2 Requirements of Specification

Requirement specification involve identifying and recording the software system's capabilities and constraints[24]. This section outlines the goals of WellTrack, including its intended functions for officers and administrators. Information related to wellness, such as stress scores, resource views, and counseling appointments, will be stored in a centralized database. The system must allow users to register with unique credentials, prioritizing secure authentication[23].

Registered users must log in to access system resources, such as personal stress history and the peer support module. The system is compatible with modern browsers and optimized for mobile devices (Android and iOS) to ensure officers can access wellness support while on duty.

3.2.0 Functional Requirements of the System

Functional requirements define the behaviors and feature the system must support. Key functionalities include:

Some of the system's key functionalities are outlined below:

- a) User Authentication and Authorization: Secure login with role-based access for Officers, Counselors, and Admins.
- b) Stress Data Collection and Management: Storing results from PHQ-9 or similar assessments, editable only by the individual or authorized medical personnel.
- c) Resource Filtering: Users can search for wellness articles by category (e.g., PTSD, Sleep, Anxiety).

- d) Real-Time Wellness Visualization: Generating personal charts displaying stress trends over time.
- e) Resource Center Integration: Access to a digital library of mental health documents and videos.
- f) Welfare Notifications: Automated alerts for upcoming counseling sessions or new wellness tips.
- g) Peer Support Forum: A secure space for officers to interact and share experiences anonymously.
- h) Counseling Management: A module for officers to request and schedule confidential sessions.
- i) User Feedback and Support: A system for reporting technical issues or providing feedback on resources.

3.2.1 UML Models

The UML Models were designed for describing object-oriented systems, and, at the architectural design stage, you often want to describe systems at a higher level of abstraction.

Unified Modeling Language (UML) is a general-purpose modelling language. The main aim of UML is to define a standard way to visualize the way a system has been designed. It is quite similar to blueprints used in other fields of engineering and most especially software engineering[25].

3.2.2 Use Case Diagram

The Use Case Diagram display how different users interact with the system and what core functionalities are provided. The primary actors in the system include the Admin and Registered Officers. The Administrator is responsible for managing the system's backend functionalities. This includes logging into the system, managing user accounts, uploading wellness data, sending manual notifications, posting and deleting wellness-related informations, and accessing an administrative dashboard for monitoring system activity and data.

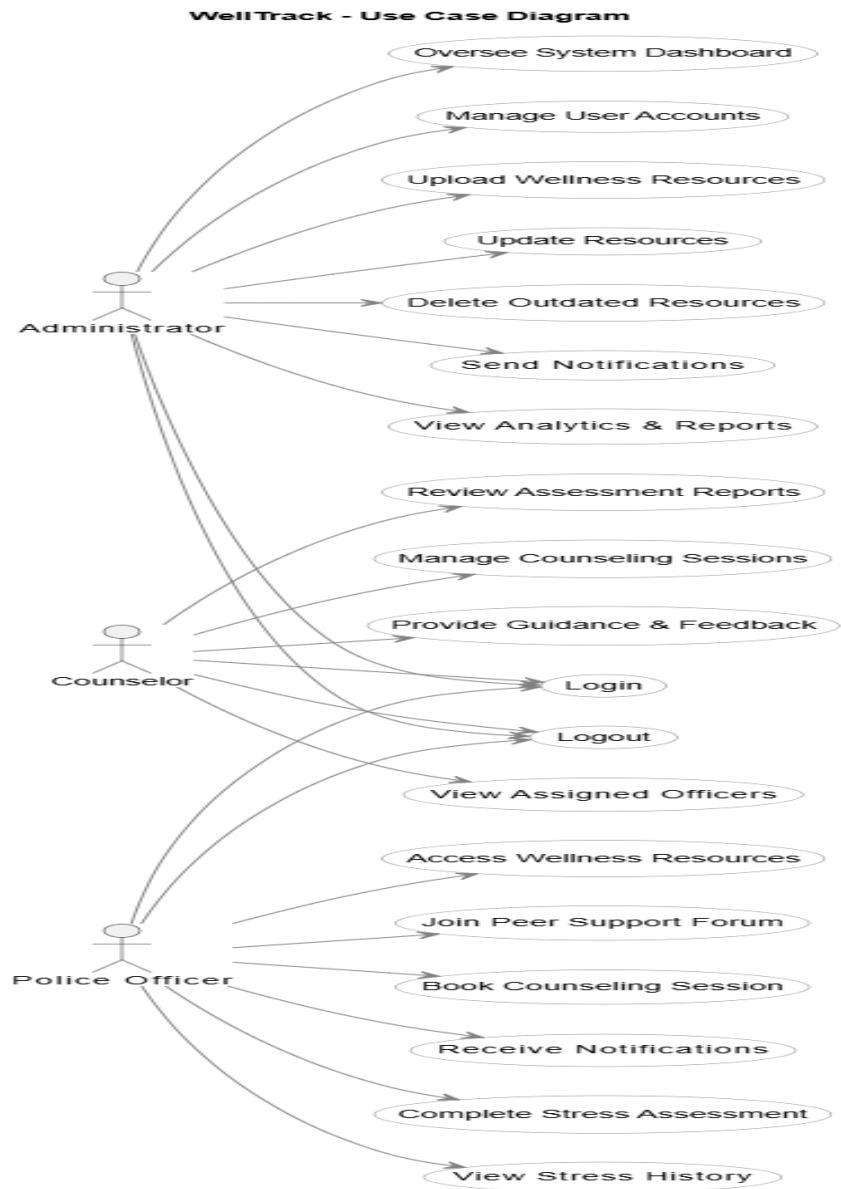


Figure 3.1: UML Use Case Diagram.

3.2.3 Entity Relations Diagram

Entity-Relationship Diagram (ERD) is essential for guaranteeing the effective arrangement, consistency, and structure of data in database design. The Web-Based wellness System's ERD is made to show how various entities work together within the system, ensuring smooth data processing and retrieval.

Visual Representation:

The ERD provides a clear visual depiction of how data is structured and connected within the system. This helps developers, system administrators, and users understand relationships between different components in an intuitive manner.

Database Design:

The ERD plays a fundamental role in defining the database schema by identifying entities (database tables), their attributes (columns), and relationships (References). This results in a well-organized, structured database that supports the system's intended functionalities.

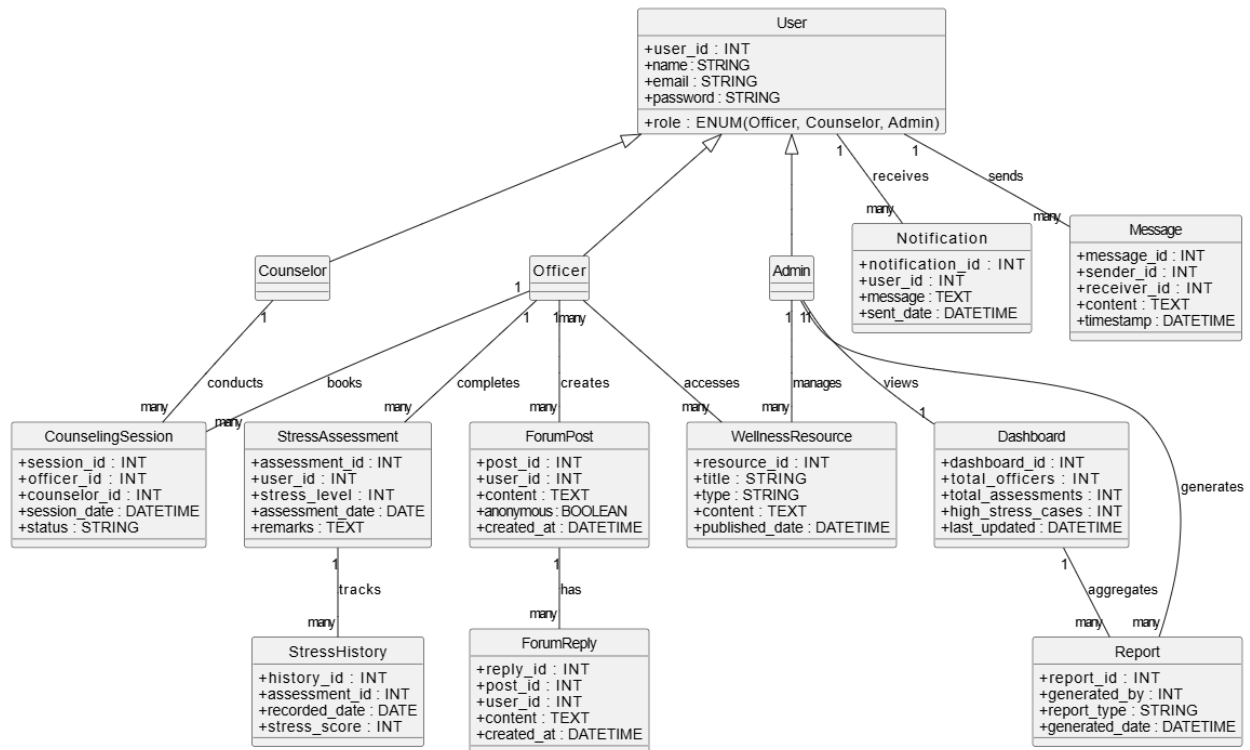


Figure 3.2: UML Class Diagram.

3.2.4 Sequence Diagram

A Sequence Diagram is a type of UML (Unified Modeling Language) diagram used to model the flow of messages or interactions over time between objects in a system. It shows how different parts of a system interact with each other to complete a specific process or use case.

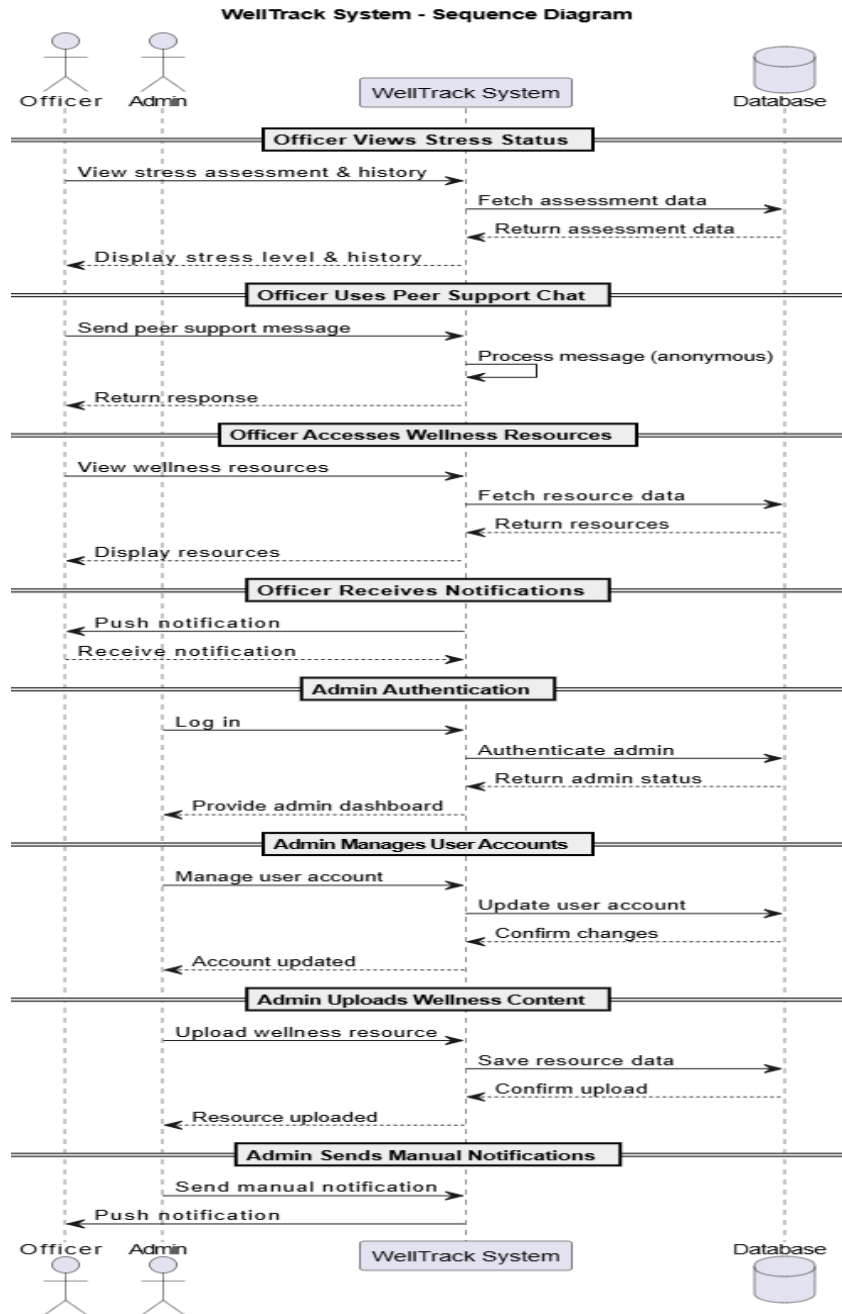


Figure 3.3: Sequence Diagram.

3.2.1.5 Activity Diagram

WellTrack - Wellness Management System (Activity Diagram)

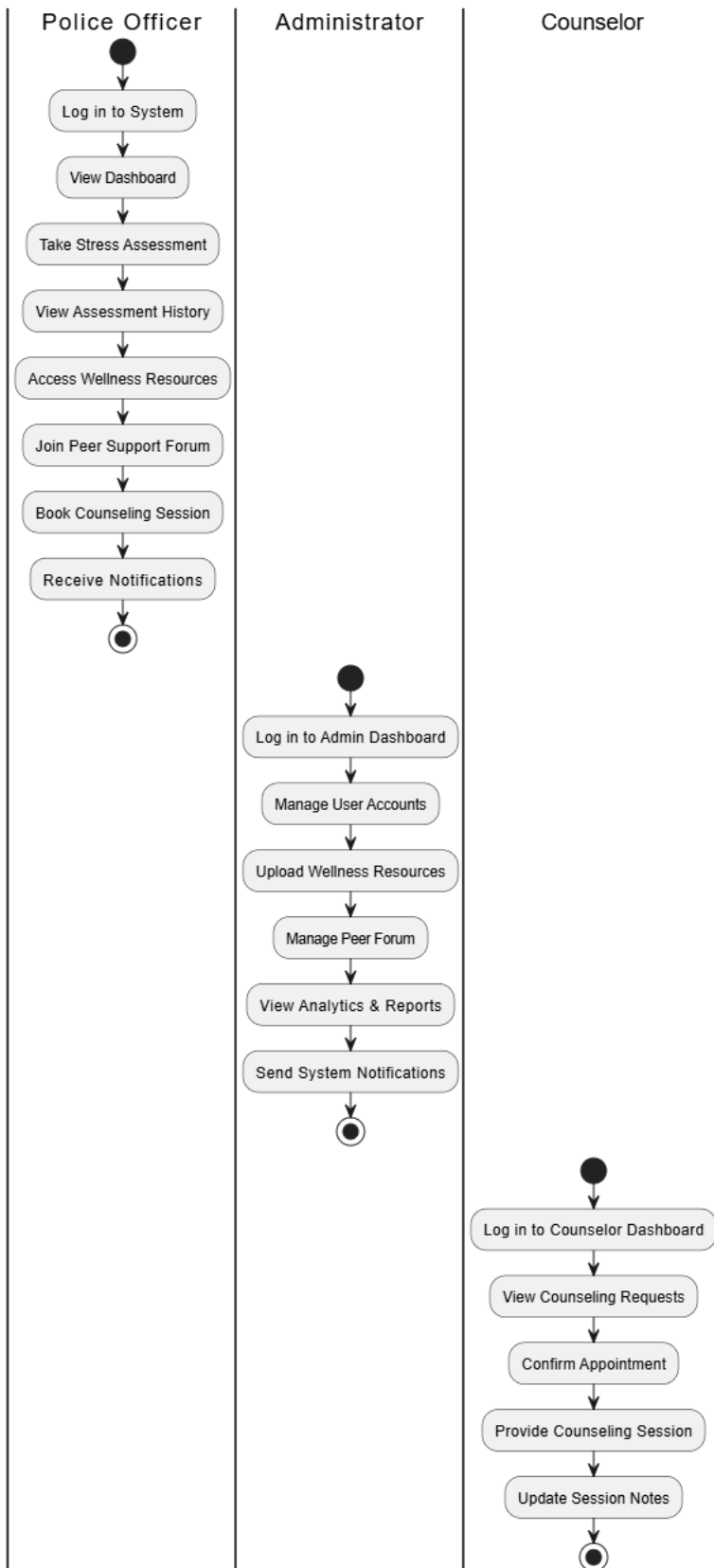


Figure 3.4: Activity Diagram

Nonfunctional Requirements of the System

The system must be user-friendly, responsive, and secure. Performance is essential, ensuring that wellness data updates within seconds[26].

Because of its intended function, the system must be able to accommodate numerous users at once and manage a large number of concurrent connections without experiencing any performance issues. Security is crucial, necessitating robust authentication methods like role-based access control to specify user permissions and multi-factor authentication for administrators. Audit logs will monitor system behavior to identify any security issues, and sensitive data, such as user credentials and wellness reports, must be encrypted to avoid unwanted access[27].

To maintain continuous service, the system must be available 24/7, with failover mechanisms in place to minimize downtime. Reliability is essential, ensuring that wellness-related data can be accessed and updated at any time. As usage increases, scalability must be prioritized, allowing the system to accommodate growing data volumes and user traffic without compromising performance. The system's modular architecture will also support future expansions and feature integrations. This system shall have the capacity for multiple login instances of different users.

The system shall also have good security features to address the protection of data and resources from unauthorized access and breaches. It involves authentication, authorization, encryption. Another essential need is maintainability, which guarantees that the system is simple to upgrade, troubleshoot, and enhance over time. While automated updates will help maintain security patches and additions current without interfering with operations, a well-organized and documented

codebase will make maintenance easier. The Web-Based Wellness System will offer a strong, scalable, and secure platform that efficiently serves officers while preserving high performance and dependability by satisfying these nonfunctional needs.

3.2.5 The Architecture of the System.

The architecture of the web-based Wellness system shows the structural design, components, interactions, and software used to build a scalable, secure, and efficient platform. This architecture ensures seamless data processing, updates, and a user-friendly experience for administrators and general officers. The system follows a three-tier architecture, which consists of the presentation layer, application layer, and database layer.

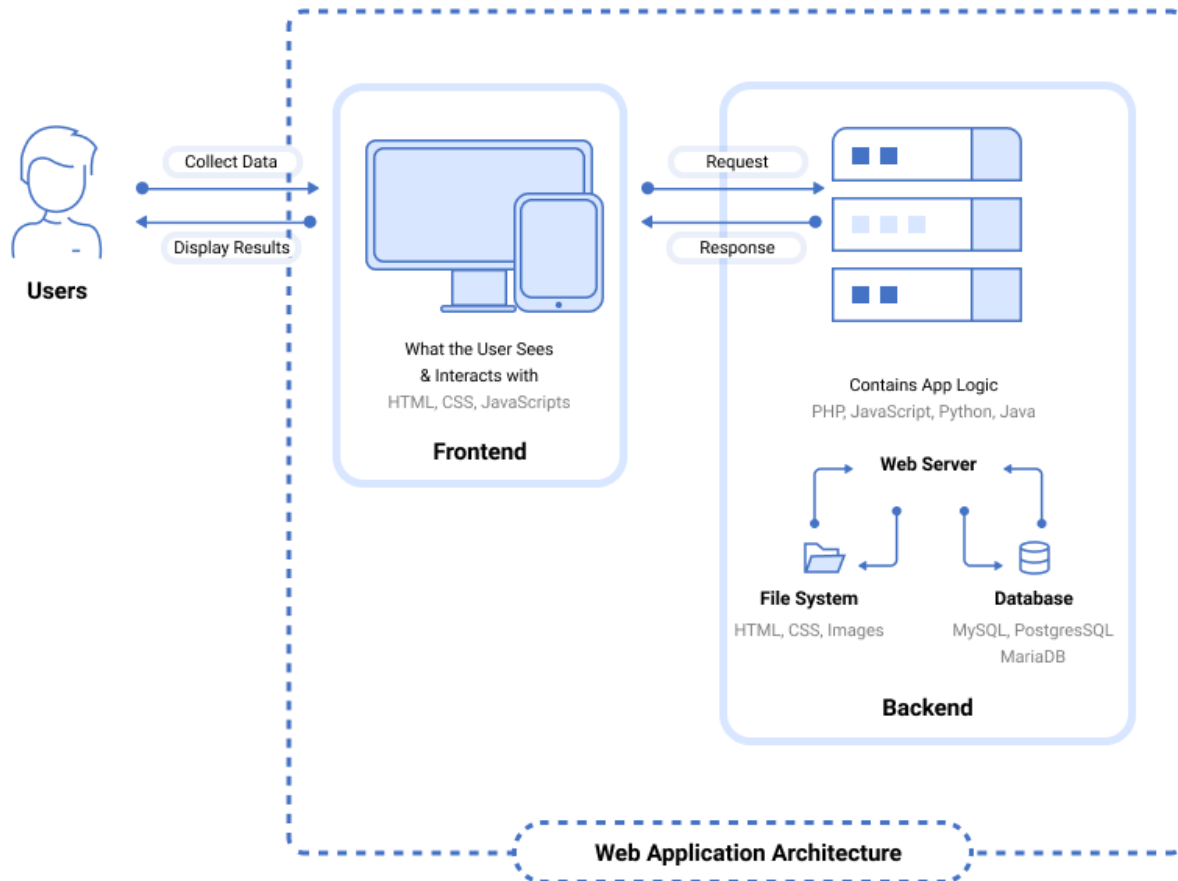


Figure 3.5: Architecture Diagram.

The presentation layer serves as the user interface, allowing users to interact with the system through a web browser. It is developed using HTML, CSS (TailwindCSS, JavaScript (Next.js) for responsive design, ensuring compatibility with various devices. This layer also integrates JavaScript frameworks to enhance user interactions, such as assessment updates and notifications.

The application layer acts as the core processing unit, handling user requests, and system functionalities. This layer is built using Nextjs which process requests from users and communicate with the database. It manages wellness data retrieval, visualization, notifications, user authentication, and real-time chat functionalities. APIs are implemented to fetch data from

global health organizations, ensuring that the system provides accurate and up-to-date wellness information[28].

The database layer is responsible for storing and managing structured wellness-related data. It utilizes a MongoDB database to efficiently handle officer records, assessment statistics, and system logs. This layer ensures data integrity, scalability, and security, implementing encryption and backup strategies to prevent data loss.



Figure 3.6: 3-tier Architecture.

3.3.0 Components Design

The Components Design is a type of software design and architecture which plays a vital role in the development lifecycle of software products as it entails the segmentation of the system into smaller, more manageable, and reusable units known as components. This stage comes after the architectural design phase and encompasses the definition of the intricate framework,

functionality, and interconnections of these components. The main objective is to convert the architectural plan into specific specifications that direct the execution of the system.

Accessibility and user interactions are the responsibility of the User Interface (UI) Component. The dashboard for tracking wellness data, the login and registration pages, the search and filtering features, and the interactive cards for viewing impacted areas are all included. Because HTML, CSS, JavaScript, and TailwindCSS were used in its development, the user interface is responsive and usable on a variety of devices[27].

The Authentication and Authorization Component manage user access control, ensuring that only authorized users can access specific system features. It supports role-based access control (RBAC), where users are classified as general users and administrators.

The Wellness Data Management Component is responsible for storing, updating, and retrieving real-time assessment data. It integrates with external health databases and APIs to pull the latest statistics on reported health assessment, recoveries, and fatalities. The data is structured and stored in the database layer MongoDB to allow quick access and efficient queries[28].

The Data Visualization Component presents Wellness information in a structured and meaningful way. It includes charts, graphs, and interactive maps that display assessment data over time.

The Notifications and Alert Component is responsible for sending real-time alerts and notifications about Health assessments, safety guidelines, and government directives. It supports email notifications, SMS alerts, and in-system push notifications to keep users informed.

The Chatbot and User Support Component provides real-time assistance to users, answering Wellness-related queries and guiding them through system functionalities.

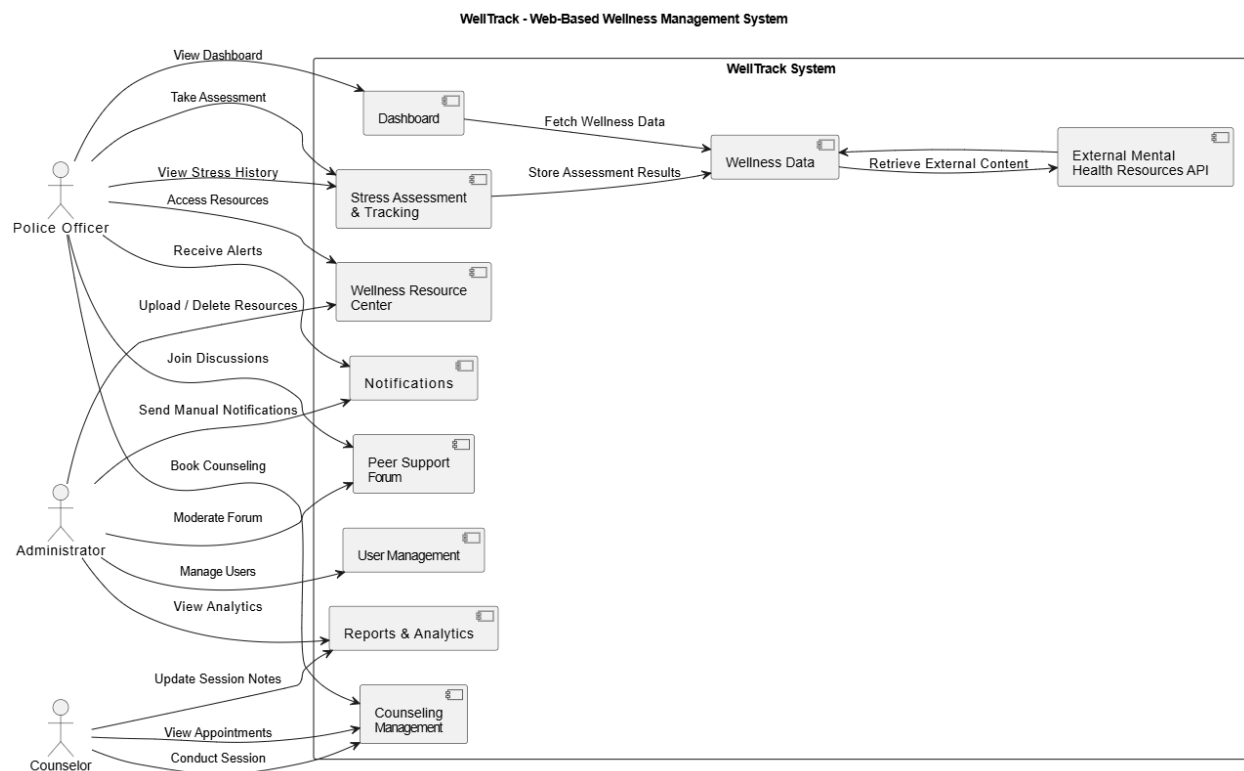


Figure 3.7: Components Design (Source: Researcher's Construction, 2024).

3.3.1 User Interface Design

The User Interface (UI) Design within the realm of software development pertains to the systematic approach of formulating the visual aspects and interactive modules of a software program that engage users. This discipline emphasizes the development of interfaces that are visually appealing, user-friendly, and easily navigable. The design and development of this repository management system highlight the principal objective of the UI design is to optimize user experience by streamlining the interaction process between the user and the system to be as effective and gratifying as achievable.

CHAPTER FOUR

IMPLEMENTATION AND TESTING

4.0 Development Tools and Platforms Consideration

The choice of development tools and platforms plays an essential role in the effective deployment and maintenance of the WellTrack wellness management system. It is imperative that these tools and platforms are able to cater to a range of functional and non-functional prerequisites of the system, including scalability, security, user-friendliness, and sustainability. This segment outlines the factors taken into account when selecting suitable development tools, encompassing front-end development, back-end development, database management, and testing[27].

In the development of the frontend, HTML5, CSS3, and the TailwindCSS framework along with JavaScript are used to design an instinctive and responsive user interface. These tools help to improve performance and allow dynamic data rendering for real-time wellness updates. For the backend, Node.js (Express.js 10+) is used due to its ability to manage server-side logic, authentication, and validation, as well as communication between the database and the frontend.

The Express framework supports real-time data processing and provides secure API endpoints to facilitate smooth data exchange. For effective data storage, a non-relational database like MongoDB is used for storing unstructured records of assessments and resources. To enable real-time updates, AJAX and WebSockets are utilized to push live notifications and alerts to officers, ensuring that wellness-related information is updated instantly[29].

JavaScript, Next.js, Node.js and MongoDB are the main tools used to develop both front and back ends of the system. The system tool provides easy means to add, edit and delete which possess certain functionality and used in implementing system software.

4.1 Front End

The front end of this system uses Next.js and CSS to develop the Graphical User Interface (GUI), which displays the dashboard, page layout and arrangements, enabling the users to easily navigate and access the system.

CSS which is also Cascading Style Sheets is a style language used for describing the look and formatting of a document written in a markup language.

JAVASCRIPT is a dynamic computer programming language which mostly used as part of web browsers, whose implementations allow client-side scripts to interact with the user, control the browser and alter document content that is displayed[23].

Back End

MongoDB database is a non-relational database management system which was used to design for the database of the system which all records are stored and being manipulated from and MongoDB atlas is a web-based tool for managing MongoDB databases which was also used for the backend development.

4.2 Database schemes

A database schema is an architecture that outlines the logical structure of the database used in the development of this system. It deals with how wellness data is organized; this is inclusive of logical constraints such as table names (e.g., users, assessments, counseling sessions), fields, data types,

and the relationships between these entities. The main role of this schema is to ensure that the database is structured efficiently to support the high confidentiality requirements of the application[29].

An effective database design plays a crucial role in minimizing data redundancy. In the context of WellTrack, the relational model was employed due to its straightforward structure and emphasis on relationships between officer roles and their corresponding data access levels.

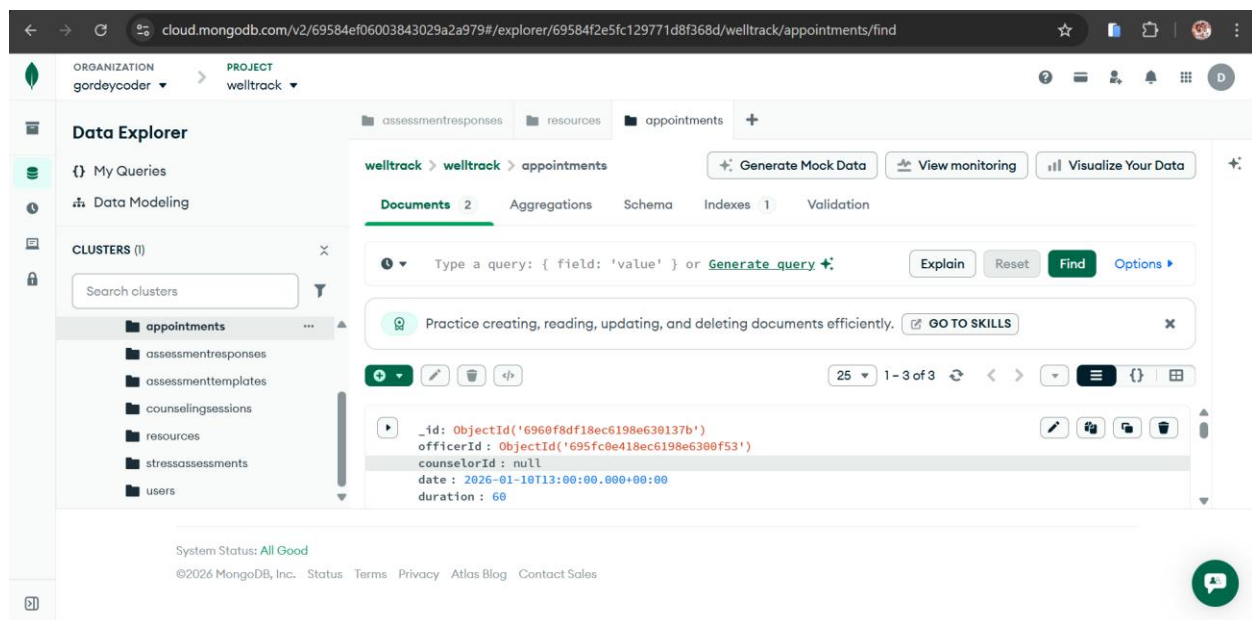


Figure 4.0: Database scheme.

4.3 User interfaces

The User Interface (UI) of this system is the interface which the officers interact with to navigate the software. It includes pages, buttons, and menus designed for productive user-friendly communication, ensuring officers can access support with minimal difficulty.

4.3.1 Landing Page

This is the first page interface, providing an introduction to WellTrack and highlighting its purpose in supporting the Ghana Police Service. It features a navigation bar with buttons like Home, About, Wellness Resources, and Login to facilitate navigation.

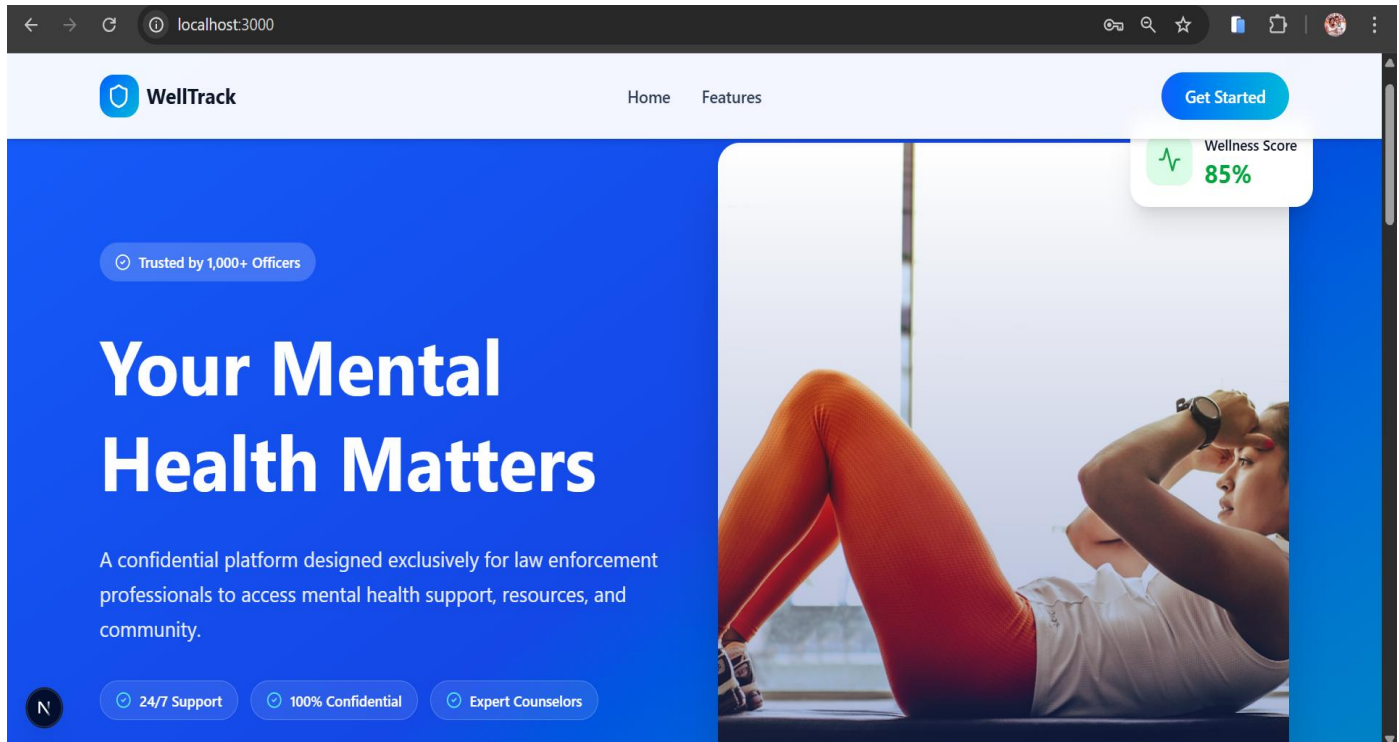


Figure 4.1: Landing Page.

The login page is a critical security component that ensures access based on user roles (Officer, Counselor, or Admin). It restricts unauthorized access to sensitive mental health data while enabling registered users to view their personal wellness trends.

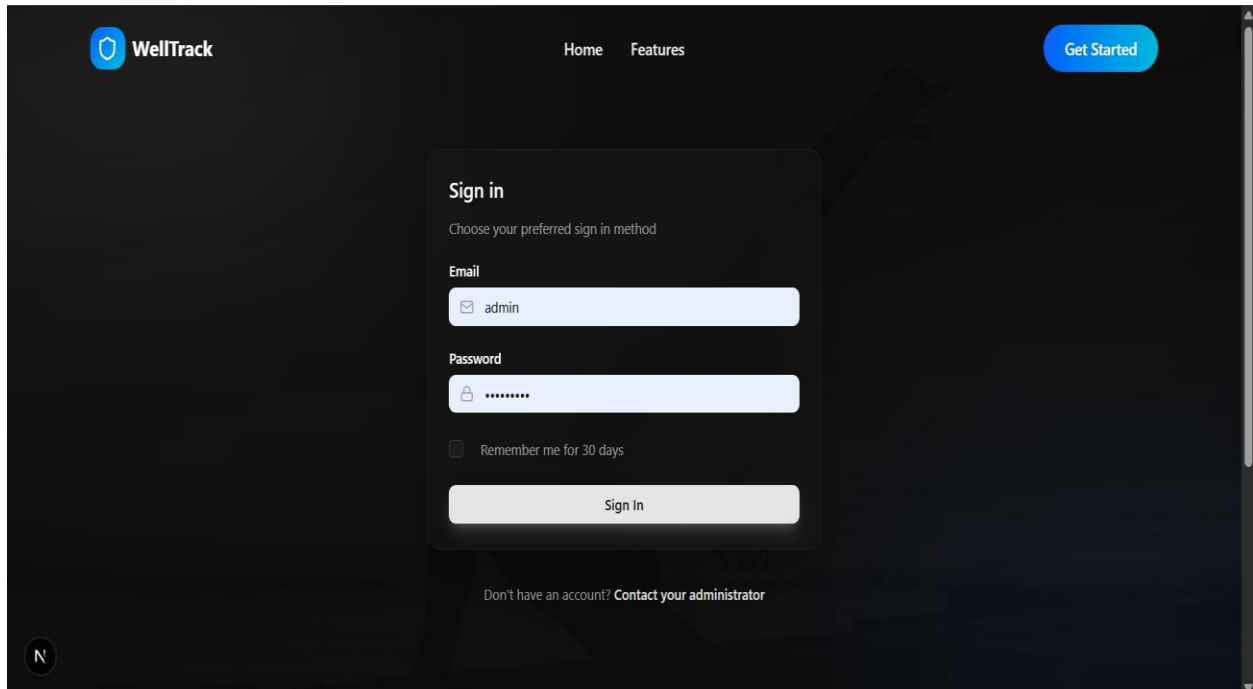


Figure 4.2: Login page.

4.3.3 About page

This part provides a summary of WellTrack, its functions, and its goal of reducing occupational stress. it allows users to understand how the system maintains their anonymity and provides support.

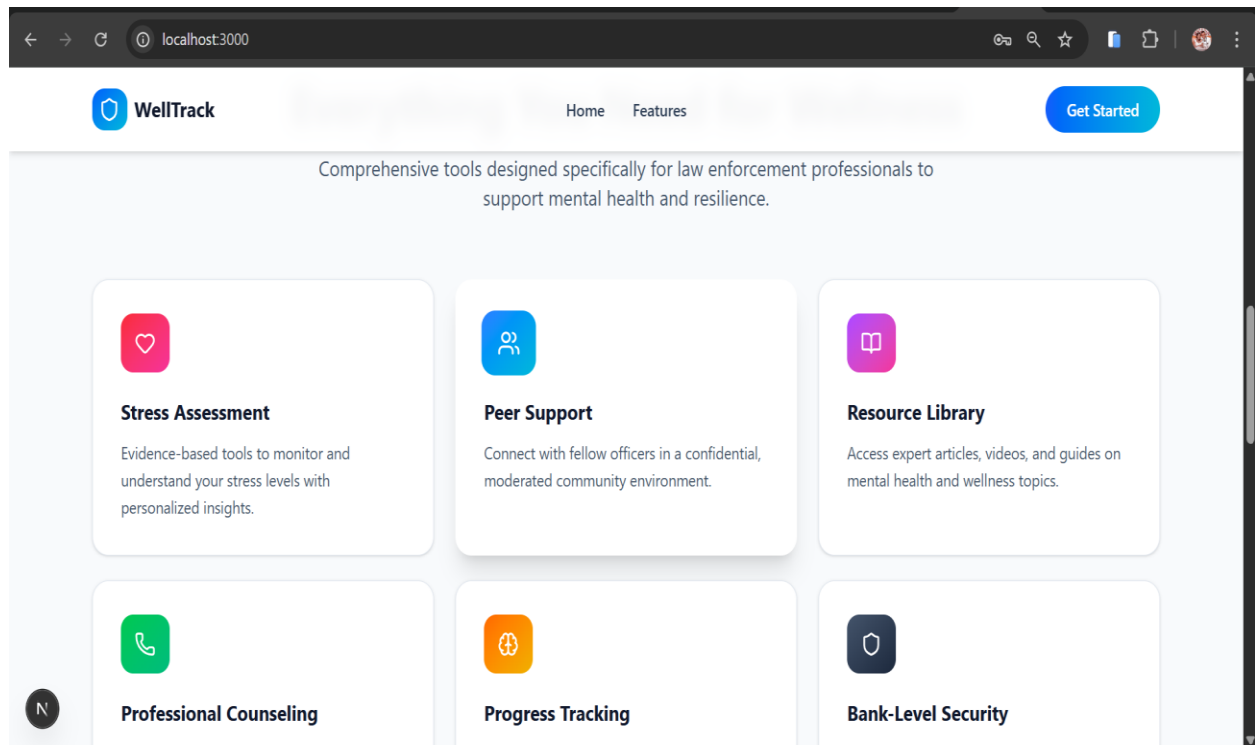


Figure 4.3: About Page.

4.3.4 Contact and Helpdesk Page

This component allows easy communication between the officer and the system administrators or welfare department. It helps users to provide feedback or report technical issues. Filling the contact form also allows users to request technical support via email.

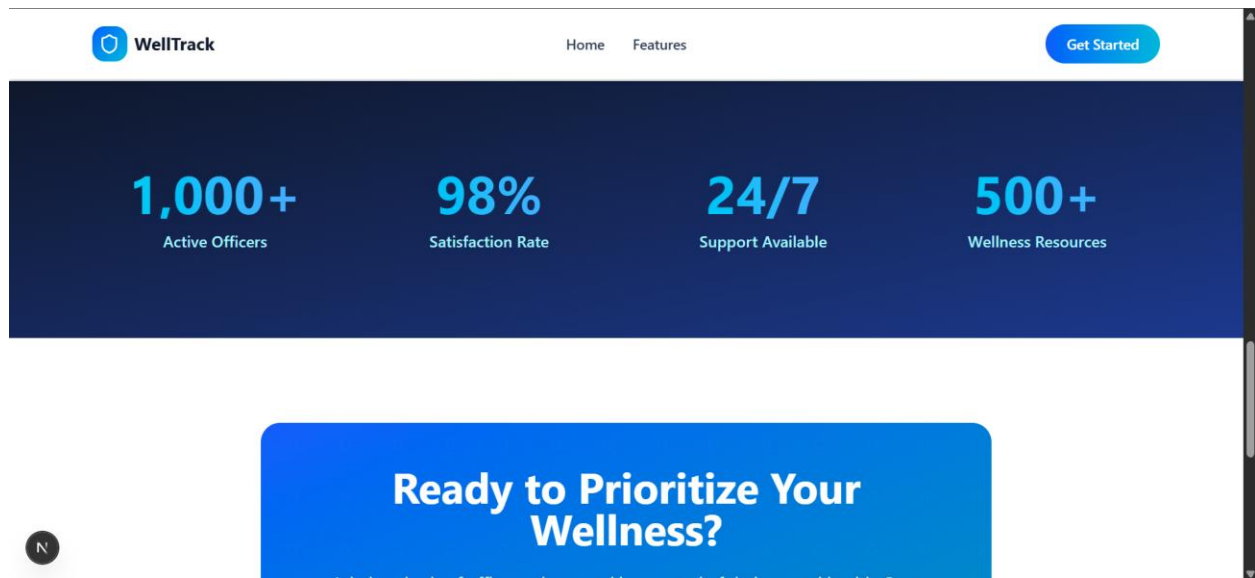


Figure 4.4: Contact page.

4.3.5 Stress Assessment (Live Update) Page

This is one of the major components of WellTrack. It provides the interface for officers to take stress tests, displaying real-time results and historical trends of their mental well-being.

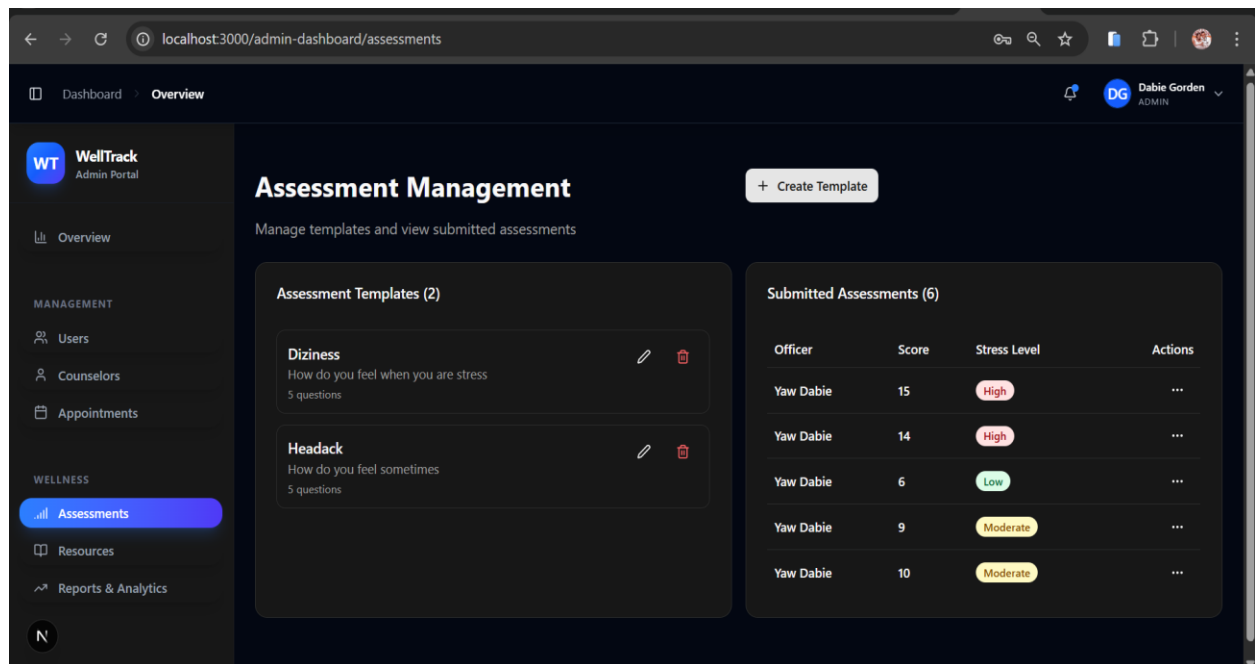


Figure 4.5: Stress Tracker page.

4.3.6 Wellness Resources

The Resources page acts as a knowledge base, displaying the latest articles on mental health, coping strategies, and public health advisories to keep officers informed on wellness matters.

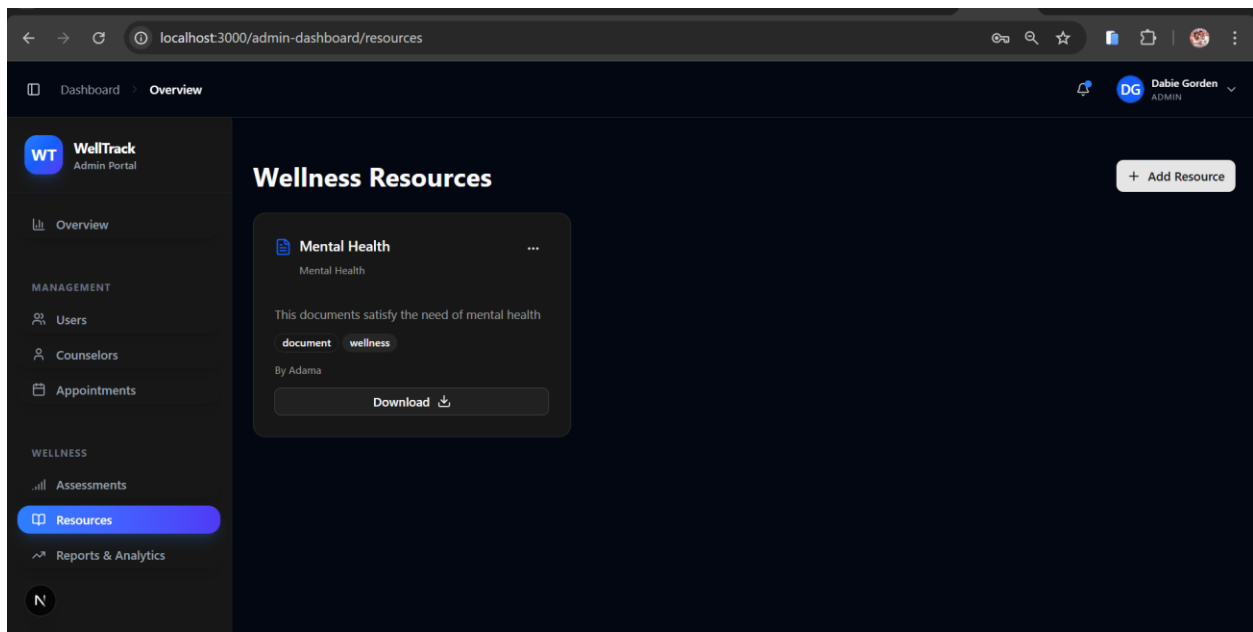


Figure 4.6: Wellness Resources page.

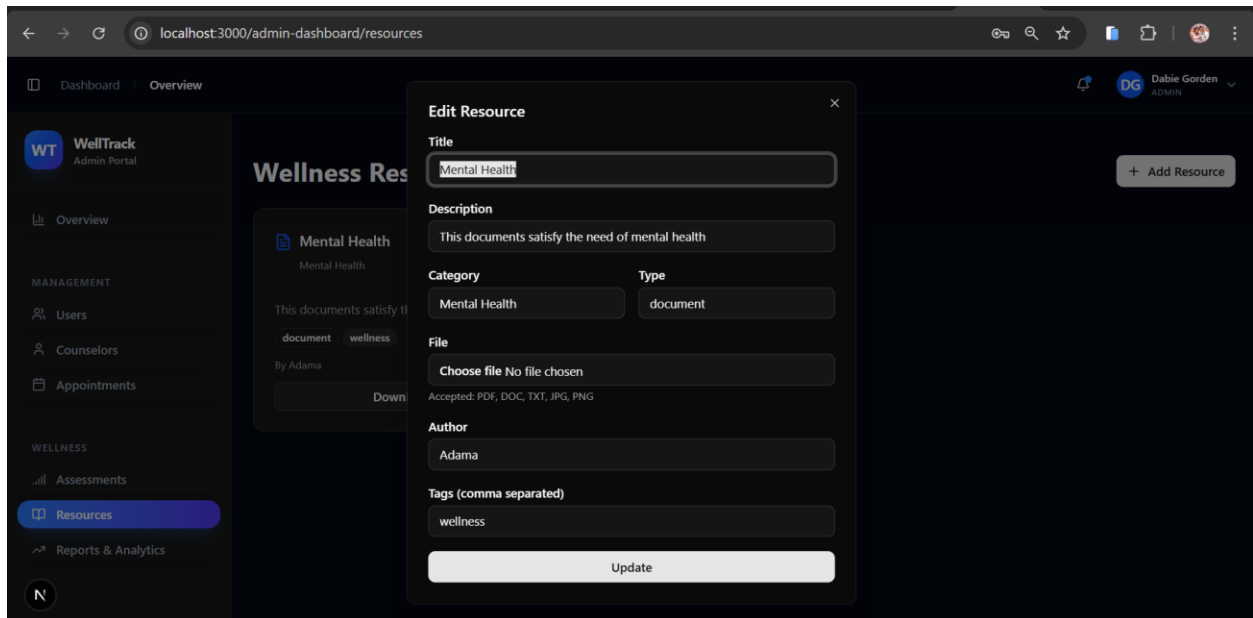


Figure 4.7: Resources Details page.

4.3.4 Dashboard Page

The Dashboard serves as the powerhouse of the system, showing an overview of an officer's wellness journey or, for admins, a graphical visualization of general wellness trends across the department to track high-risk units.

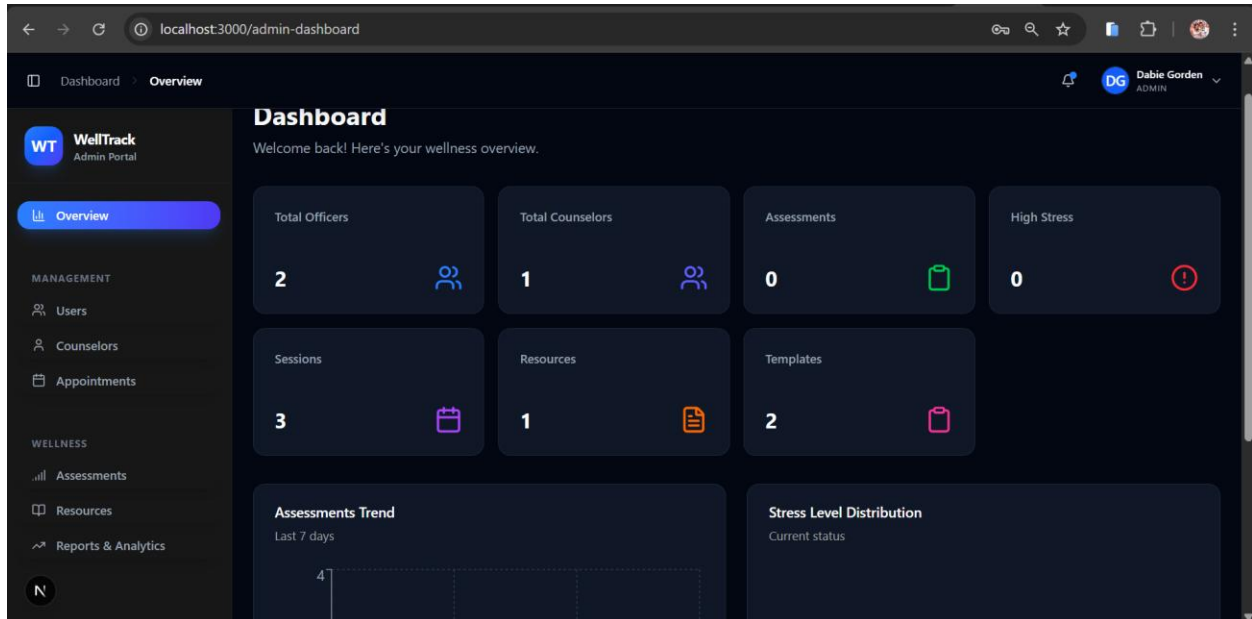


Figure 4.8: Dashboard page.

4.4 Testing

The process of testing plays a pivotal role within the software development lifecycle, that ensures the software application meets the required system specification and functions correctly. This phase follows the development stage and involves a thorough evaluation of the software to detect and resolve any defects or issues. The main goal is to verify that the application functions as intended and meets user requirements. This system after developed was tested by a few sample users by registering and login into the system, and they were able to receive notifications on their email, send feedbacks to the admins their dashboards. This was functioning efficiently and responding to user requests.

This process encompasses multiple test types are performed, including unit testing, integration testing, system testing, and acceptance testing, to validate different aspects of the software.

4.4.1 Units/ Components Testing

A unit test is a type of software test that focuses on testing individual components of a software product. The ‘units’ in a unit test can be functions, procedures, methods, objects, or other entities in an application’s source code. The main goal of unit testing is to ensure that each unit of the software performs as intended and meets requirements. Unit tests help make sure that software is working as expected before it is released. Component testing, where several individual units are integrated to create composite components. Component testing should focus on testing the component interfaces that provide access to the component functions.

4.4.2 Integration and System Testing

Integration testing is when performing tests on multiple different modules and components together. The purpose of this testing is to examine how those various modules interact with each other[30]. Firstly, every module is tested separately to make sure it works smoothly and afterward, the integration test is executed to check the performance of those modules collectively. Even if the units of software are working fine individually, there is a need to find out if the units if integrated together would also work without errors. System testing, where some or all of the components in a system are integrated and the system is tested as a whole.

After the development of the software, it was tested by the users and the results is better and was functioning robustly faster than other earlier systems in the review stage of this project because this system has a small search engine functionality that responds quickly to key words typed in the search button for pandemic related news. System testing should focus on testing component interactions.

4.4.3 Deployment Model and Platforms

1. Operating system: Windows 7 or above, UNIX or MAC.
2. Processor: 2.0 GHz or higher.
3. RAM: 256Mb or more.

Server side:

1. Operating system: Windows ME or 9x/xp.
2. Processor: 2.0 GHz or higher.
3. RAM: 256Mb or more.
4. Hard Drive: 10GB

4.4.4 Acceptance Testing

Acceptance testing is the final round where the system is evaluated by the end-users (Police Welfare Officers) to decide if it fulfills their requirements and is ready for full-force deployment.[31]. This ensures the system is ready to be moved from the development environment to the live institutional environment.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATION

5.0 Discussion

The main objective of the **WellTrack** wellness management system is to improve the tracking and response to occupational stress and mental health challenges among police officers by centralizing wellness data in a real-time and accessible platform. The implementation of this software helps the Police Welfare Department monitor the emotional and psychological well-being of the force based on available assessment data[27].

Role-based access control is implemented to ensure the high level of confidentiality required for sensitive health data, allowing only authorized users like counselors and administrators to have access to specific system functionalities. The roles are divided into Officers and Administrators (Welfare Officers/Counselors); this structure preserves the integrity of sensitive data while enhancing the availability of wellness-related resources and support[32].

The results of this system after user acceptance testing (UAT) was conducted by a sample group of officers were highly positive, satisfying user specifications regarding anonymity and ease of use. Effective management of wellness data, including stress scores, resource views, and

counseling appointments, is a core strength of the system. To improve data retrievability, WellTrack uses categorized modules and metadata. Officers can swiftly find specific wellness articles, tracking history, or support contacts with the help of the advanced search feature.

The system uses a scalable architecture that allows for increasing volumes of data from various police stations across the country. The scalability ensures that the platform remains responsive even as the number of active users grows

[31] Performance optimization strategies, such as server-side caching and asynchronous data loading, ensure that officers can access support quickly, even in high-pressure situations.

5.1 Discussion of Achievements

The introduction of the WellTrack system marks a significant milestone in the proactive management of police welfare. It has transformed how officers interact with mental health resources, providing a secure and efficient platform. The following have been achieved:

- a) **Enhanced Organization and Accessibility:** The system has centralized all wellness information, including stress assessments, resource libraries, and counseling schedules. This has streamlined the process of seeking help and reduced the stigma associated with physical visits to welfare offices.
- b) **Real-Time Notification and Alerts:** An automated notification feature informs officers about upcoming counseling sessions or high-stress alerts based on their assessments. This ensures timely intervention and better situational awareness for the Welfare Department.

- c) **Data-Driven Wellness Analytics:** The system's ability to visualize stress trends is a major accomplishment. It allows administrators to identify units or regions under extreme pressure, enabling proactive resource deployment and policy adjustments.
- d) **Centralized Wellness Resource Dissemination:** A resource section allows administrators to manage and upload verified mental health articles and videos. This ensures officers receive accurate information directly from reliable professional sources.
- e) **Secure and Anonymous Access:** The implementation of stringent access controls ensures that an officer's mental health status is protected. While administrators see force-wide trends, individual privacy is maintained according to health data protection rules.

5.2 Conclusion

The development of the WellTrack wellness management system can profoundly impact the law enforcement environment in Ghana. By providing a centralized, organized platform, the system ensures that the analysis and response to occupational stress are handled effectively and efficiently.

This system gives welfare professionals immediate access to force-wide wellness trends, facilitating prompt interventions. The impact of burnout and PTSD is lessened by integrating data-driven tracking with early warning systems. For the general officers, WellTrack offers a structured approach to access timely resources, reducing the "silent struggle" and increasing awareness of mental health. Due to the user-friendly interface, officers of all technical backgrounds can engage with the software effectively.

Ultimately, this system contributes to a more resilient police force by improving the timeliness, accuracy, and accessibility of wellness information, thereby supporting better preparedness and stronger health outcomes for those who serve the public.

5.3 Recommendations

- a) **Continuous Training and Support:** To ensure effective utilization, training programs must be established for welfare officers and unit commanders. Regular workshops on digital wellness tools can enhance user engagement and trust in the system.
- b) **Enhanced Data Integration:** Future versions should prioritize integrating with existing police medical records while maintaining strict privacy. Advanced natural language processing could be used to analyse anonymous feedback for better sentiment tracking.
- c) **Mobile Accessibility:** Extending full WellTrack capability to a dedicated mobile app is critical for field officers. Mobile applications that allow offline assessment data gathering in remote areas would significantly improve coverage.
- d) **Automated Backup and Resilience:** Implementing redundant data recovery systems is essential during institutional crises. Regularly scheduled backups will protect against data loss and ensure continuity of welfare services.
- e) **Enhanced Security Measures:** Given the sensitivity of mental health data, security is paramount. Implementing multi-factor authentication (MFA) for all users and conducting regular security audits will help maintain the trust of the officers.

- f) **Scalability Monitoring:** As the entire Ghana Police Service adopts the platform, the architecture must be supported by cloud services to ensure high performance under heavy load.
- g) **Regular Updates:** The user interface should be refreshed regularly based on officer feedback to ensure it remains a supportive and non-intimidating environment.

5.4 Future Prospects

Building on the success of WellTrack, we plan to expand its capabilities. Future enhancements include integrating more sophisticated AI-driven chatbots for immediate crisis intervention and suicide prevention. This will allow for 24/7 immediate support when human counselors are unavailable.

Another significant prospect is the addition of "Peer Support Matching," where the system intelligently connects officers with similar experiences for anonymous peer-to-peer mentoring. Finally, we anticipate enhancing the system's ability to integrate with wearable fitness trackers to monitor physiological stress indicators (like heart rate) in real-time, providing an even more comprehensive wellness profile for the modern police officer.

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APPENDIX

HOW TO DEPLOY THE SYSTEM

Step 1: Install Visual Studio Code IDE, especially the latest version of the IDE (Visual Studio Code version 1.100) by downloading it from the official website.



Figure: Downloading the IDE.

Step 2: Install PHP, a web server (WAMP OR XAMP), MySQL and PhpMyAdmin.

XAMPP Control Panel v3.3.0

Modules

Service	Module	PID(s)	Port(s)	Actions
☐	Apache			Start Admin Config Logs
☐	MySQL			Start Admin Config Logs
☐	FileZilla			Start Admin Config Logs
☐	Mercury			Start Admin Config Logs
☐	Tomcat			Start Admin Config Logs

Config

Netstat

Shell

Explorer

Services

Help

Quit

```

11:49:10 AM [main] Initializing Control Panel
11:49:10 AM [main] Windows Version: Enterprise 64-bit
11:49:10 AM [main] XAMPP Version: 8.2.12
11:49:10 AM [main] Control Panel Version: 3.3.0 [ Compiled: Apr 6th 2021 ]
11:49:10 AM [main] You are not running with administrator rights! This will work for
11:49:10 AM [main] most application stuff but whenever you do something with services
11:49:10 AM [main] there will be a security dialogue or things will break! So think
11:49:10 AM [main] about running this application with administrator rights!
11:49:10 AM [main] XAMPP Installation Directory: "c:\xampp\"
11:49:10 AM [main] Checking for prerequisites
11:49:10 AM [main] All prerequisites found
11:49:10 AM [main] Initializing Modules
11:49:10 AM [main] Starting Check-Timer
11:49:10 AM [main] Control Panel Ready

```

Figure: Installing the Xampp sever.

Step 3. Extract Xamp.Zip onto the base directory of your hard drive.

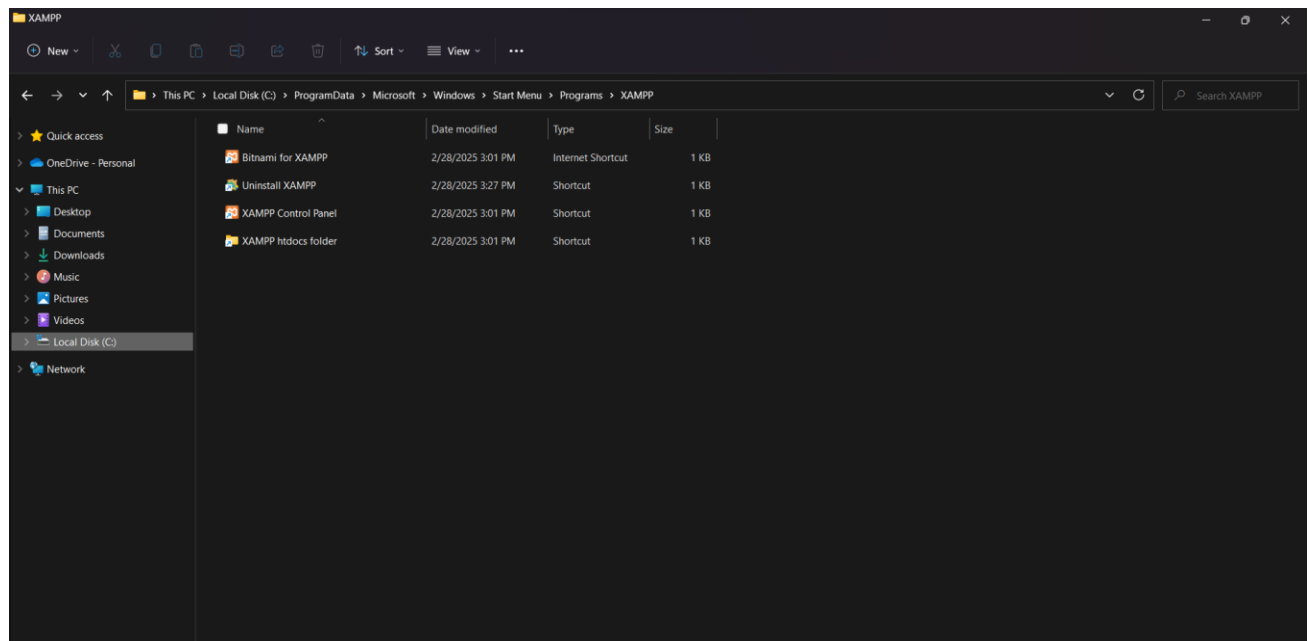


Figure: Extracting the zip folder.

Step 4: Open the extracted folder using Apache NetBeans IDE or Visual Studio Code or any text editor.

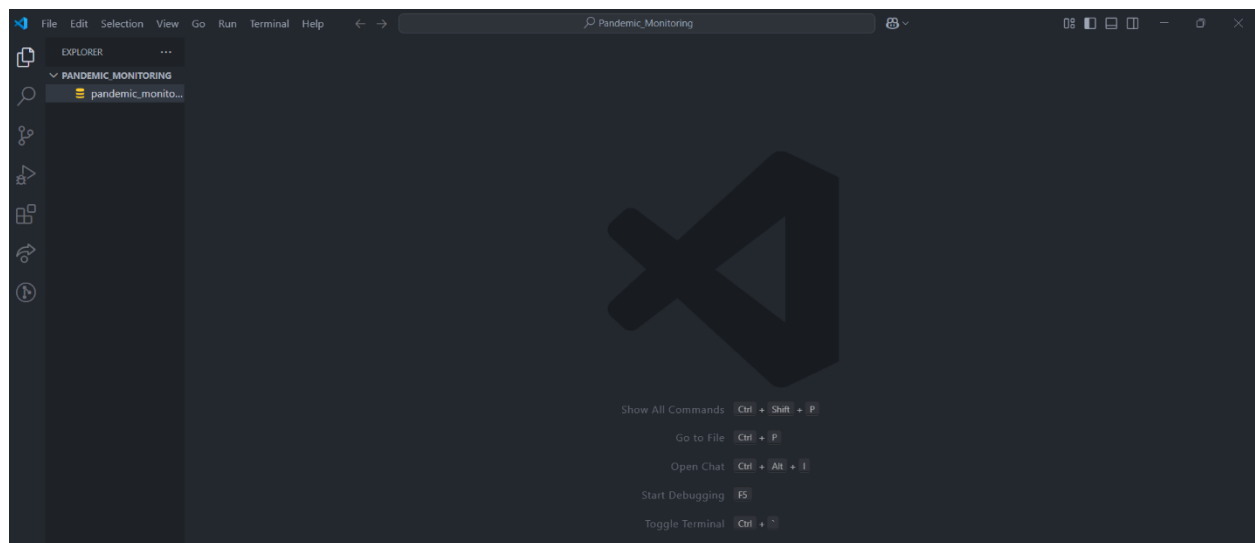
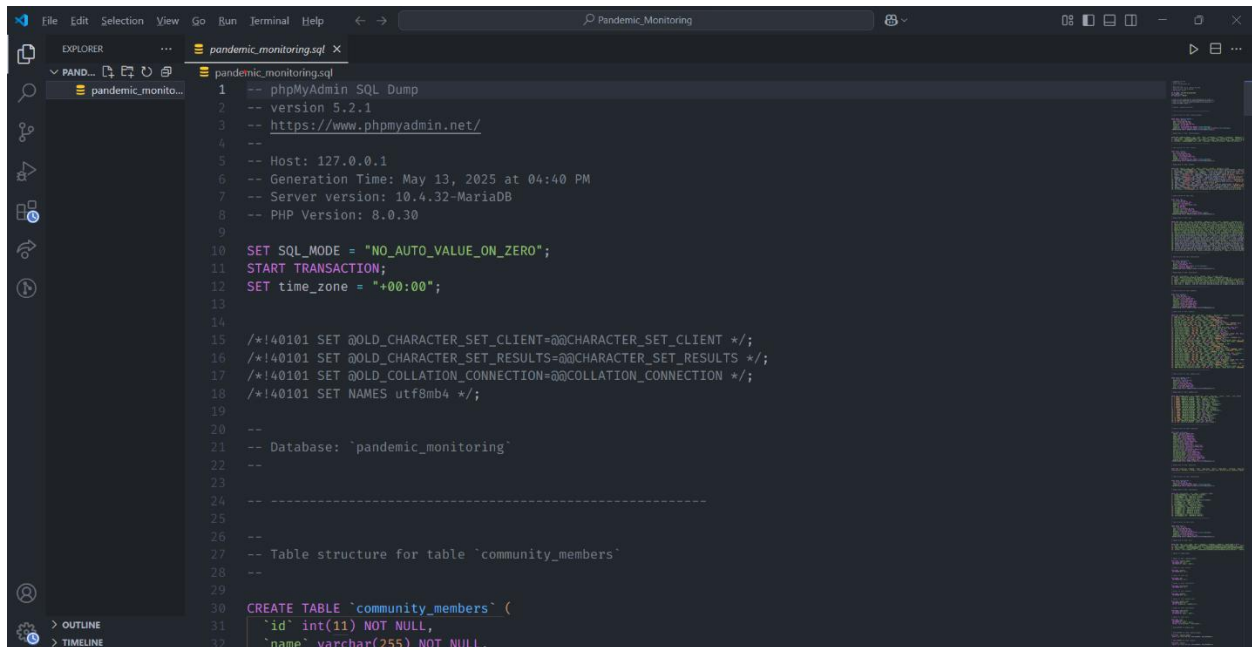


Figure: Open extracted file in vs code.

Step 5: Create a database with the name pandemic_monitoring.sql using phpMyAdmin or type <http://localhost/phpmyadmin/> on the URL of google chrome browser or Microsoft Edge or any browser.



```
1 -- phpMyAdmin SQL Dump
2 -- version 5.2.1
3 -- https://www.phpmyadmin.net/
4 --
5 -- Host: 127.0.0.1
6 -- Generation Time: May 13, 2025 at 04:40 PM
7 -- Server version: 10.4.32-MariaDB
8 -- PHP Version: 8.0.30
9
10 SET SQL_MODE = "NO_AUTO_VALUE_ON_ZERO";
11 START TRANSACTION;
12 SET time_zone = "+00:00";
13
14
15 /*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
16 /*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
17 /*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
18 /*!40101 SET NAMES utf8mb4 */;
19
20 --
21 -- Database: `pandemic_monitoring`
22 --
23 --
24 -- -----
25 --
26 --
27 -- Table structure for table `community_members`
28 --
29
30 CREATE TABLE `community_members` (
31   `id` int(11) NOT NULL,
32   `name` varchar(255) NOT NULL,
```

Figure: Create a database with the name pandemic_monitoring.

Step 6: Open your browser and enter the URL address:

<http://localhost:5000>

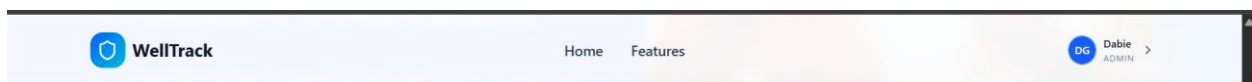


Figure: Open project in the browser.

Step7: Enter dabiegorden49@gmail.com and “12345678” in the email input and the password input of the login page respectively to open the admin panel/page of the system.

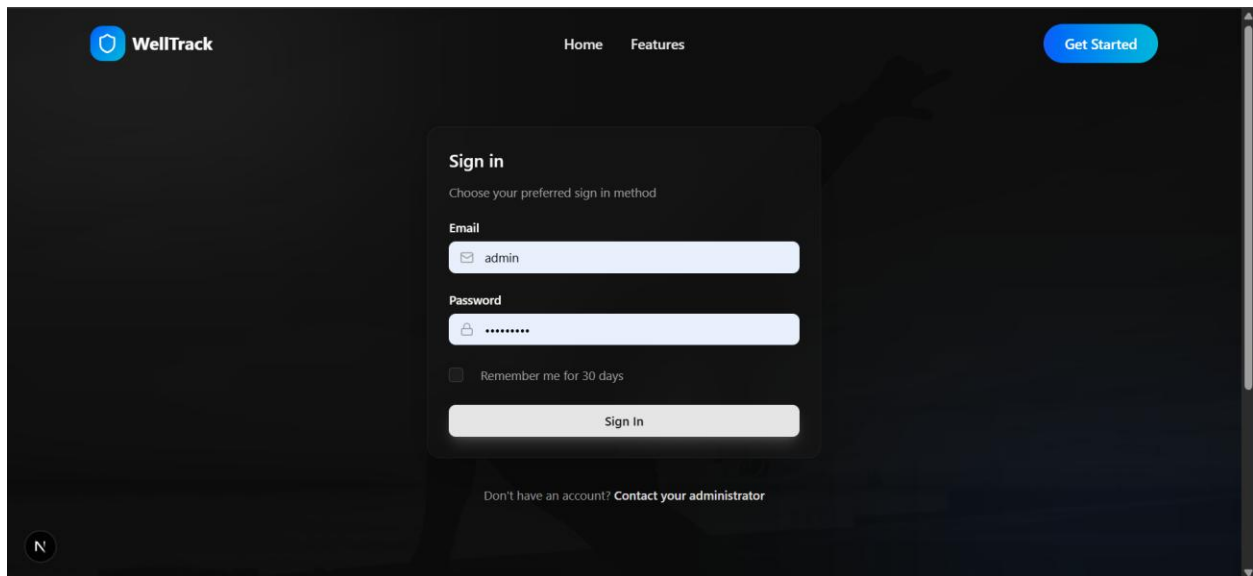
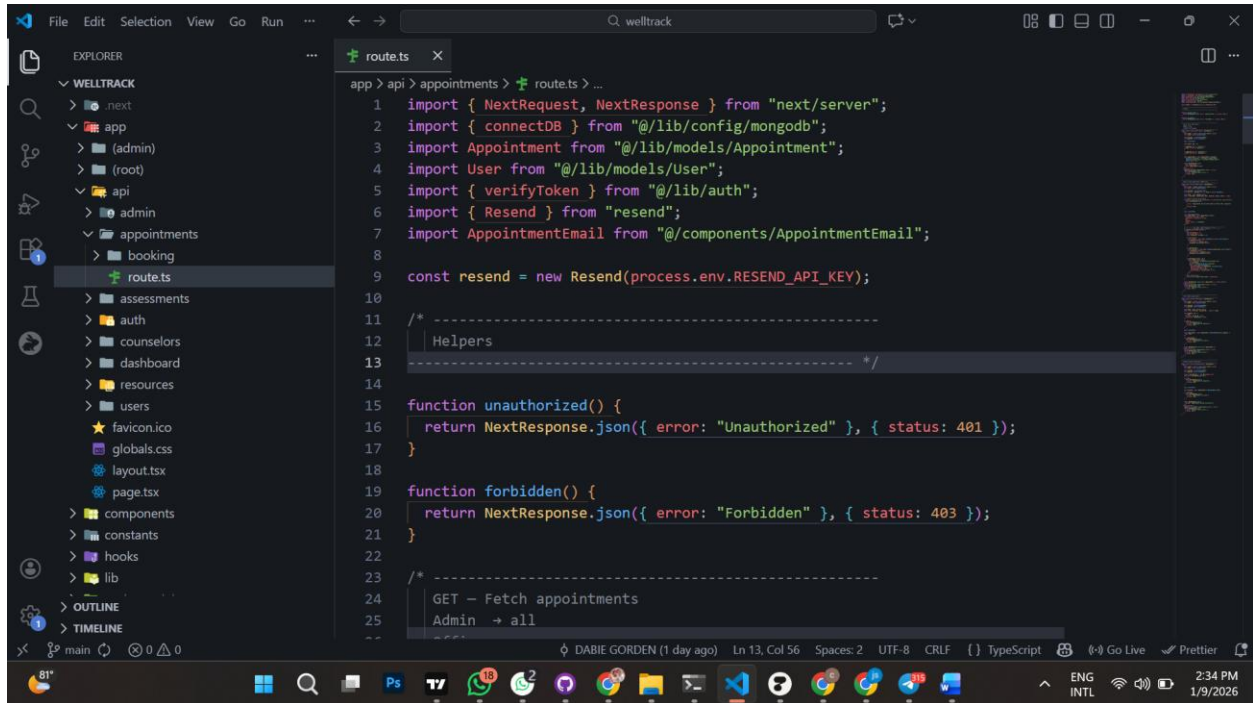


Figure: Create an account.

APPENDIX

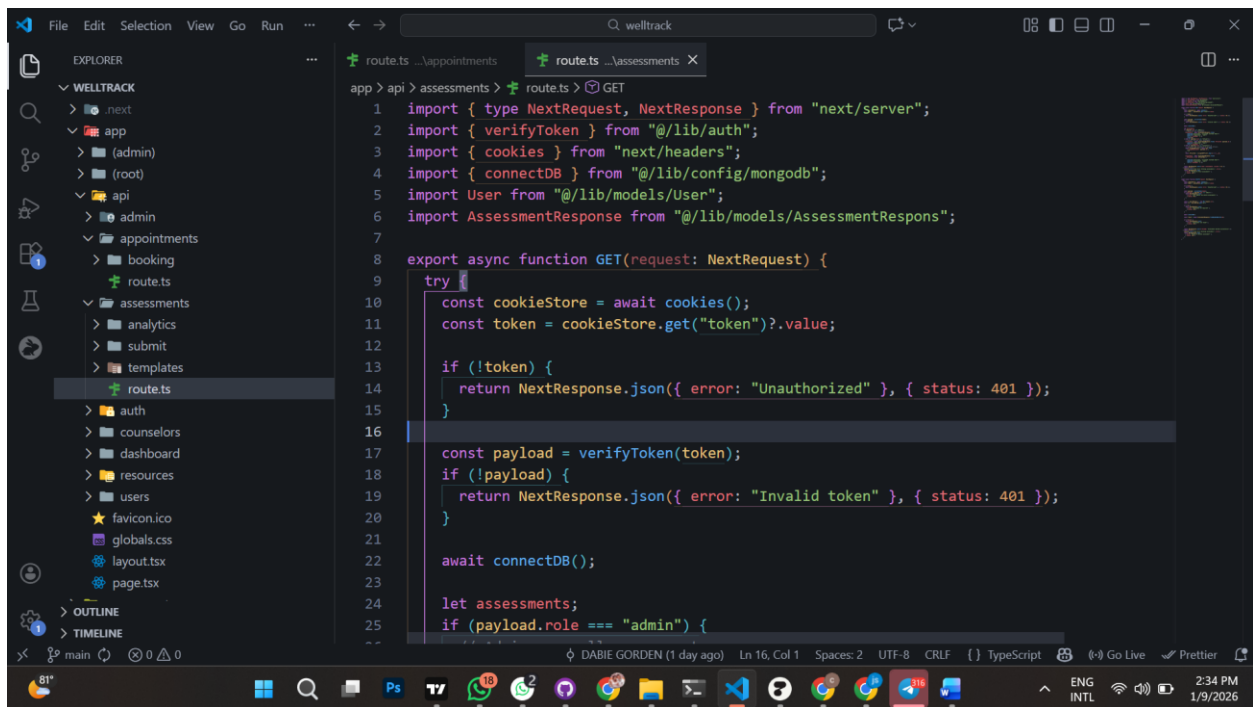
SOURCE CODE

Below are some screenshots of some of the pages of the source code of the system.



```
File Edit Selection View Go Run ... < > welltrack
EXPLORER
WELLTRACK
  > .next
  > app
    > (admin)
    > (root)
  > api
    > admin
    > appointments
      > booking
      > route.ts
    > assessments
    > auth
    > counselors
    > dashboard
    > resources
    > users
  > favicon.ico
  > globals.css
  > layout.tsx
  > page.tsx
  > components
  > constants
  > hooks
  > lib
  > OUTLINE
  > TIMELINE
main 0 0 0
61°
DABIE GORDEN (1 day ago) Ln 13, Col 56 Spaces: 2 UTF-8 CRLF {} TypeScript Go Live Prettier
1 import { NextRequest, NextResponse } from "next/server";
2 import { connectDB } from "@lib/config/mongodb";
3 import Appointment from "@lib/models/Appointment";
4 import User from "@lib/models/User";
5 import { verifyToken } from "@lib/auth";
6 import { Resend } from "resend";
7 import AppointmentEmail from "@components/AppointmentEmail";
8
9 const resend = new Resend(process.env.RESEND_API_KEY);
10
11 /* -----
12   Helpers
13   ----- */
14
15 function unauthorized() {
16   return NextResponse.json({ error: "Unauthorized" }, { status: 401 });
17 }
18
19 function forbidden() {
20   return NextResponse.json({ error: "Forbidden" }, { status: 403 });
21 }
22
23 /* -----
24   GET - Fetch appointments
25   Admin → all
26   ----- */
```

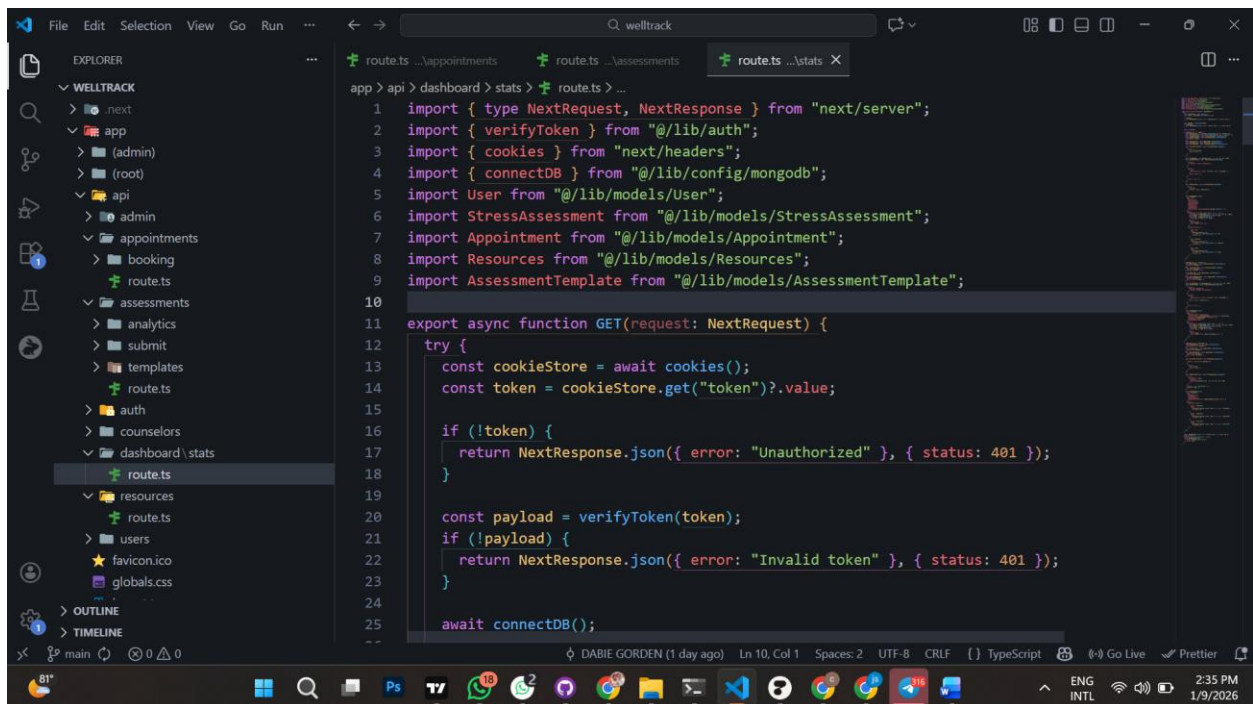
Figure 5.1: Sample source code for the contact.js – Contact Page



The screenshot shows the Visual Studio Code editor with the Explorer sidebar on the left. The project structure includes a 'WELLTRACK' folder with subfolders like 'api', 'assessments', 'auth', 'counselors', 'dashboard', 'resources', and 'users'. The 'assessments' folder is expanded, showing 'route.ts' as the selected file. The main editor displays the source code for 'route.ts', which is a Next.js GET handler. It imports necessary modules from 'next/server', '@lib/auth', 'next/headers', '@lib/config/mongodb', '@lib/models/User', and '@lib/models/AssessmentResponses'. The code defines an async function 'GET' that checks for a token in cookies, verifies it, and then connects to a database to fetch assessments for an 'admin' user. The status bar at the bottom indicates the file is 'route.ts' in the 'assessments' directory, with line 16, column 1, and a TypeScript language mode.

```
1 import { type NextRequest, NextResponse } from "next/server";
2 import { verifyToken } from "@lib/auth";
3 import { cookies } from "next/headers";
4 import { connectDB } from "@lib/config/mongodb";
5 import User from "@lib/models/User";
6 import AssessmentResponse from "@lib/models/AssessmentResponses";
7
8 export async function GET(request: NextRequest) {
9   try {
10     const cookieStore = await cookies();
11     const token = cookieStore.get("token")?.value;
12
13     if (!token) {
14       return NextResponse.json({ error: "Unauthorized" }, { status: 401 });
15     }
16
17     const payload = verifyToken(token);
18     if (!payload) {
19       return NextResponse.json({ error: "Invalid token" }, { status: 401 });
20     }
21
22     await connectDB();
23
24     let assessments;
25     if (payload.role === "admin") {
```

Figure 5.2: Sample source code for the page.jsx – News Page



The screenshot shows the Visual Studio Code editor with the Explorer sidebar on the left. The project structure is similar to Figure 5.2, but the 'dashboard' folder is expanded, showing 'stats' as a subfolder. The 'stats' folder is expanded, and 'route.ts' is selected. The main editor displays the source code for 'route.ts', which is a Next.js GET handler. It imports necessary modules from 'next/server', '@lib/auth', 'next/headers', '@lib/config/mongodb', '@lib/models/User', '@lib/models/StressAssessment', '@lib/models/Appointment', '@lib/models/Resources', and '@lib/models/AssessmentTemplate'. The code defines an async function 'GET' that checks for a token in cookies, verifies it, and then connects to a database. The status bar at the bottom indicates the file is 'route.ts' in the 'dashboard/stats' directory, with line 10, column 1, and a TypeScript language mode.

```
1 import { type NextRequest, NextResponse } from "next/server";
2 import { verifyToken } from "@lib/auth";
3 import { cookies } from "next/headers";
4 import { connectDB } from "@lib/config/mongodb";
5 import User from "@lib/models/User";
6 import StressAssessment from "@lib/models/StressAssessment";
7 import Appointment from "@lib/models/Appointment";
8 import Resources from "@lib/models/Resources";
9 import AssessmentTemplate from "@lib/models/AssessmentTemplate";
10
11 export async function GET(request: NextRequest) {
12   try {
13     const cookieStore = await cookies();
14     const token = cookieStore.get("token")?.value;
15
16     if (!token) {
17       return NextResponse.json({ error: "Unauthorized" }, { status: 401 });
18     }
19
20     const payload = verifyToken(token);
21     if (!payload) {
22       return NextResponse.json({ error: "Invalid token" }, { status: 401 });
23     }
24
25     await connectDB();
```

Figure 5.3: Sample source code for the pandemic.js

```
File Edit Selection View Go Run ...  
Q welltrack  
EXPLORER  
WELLTRACK  
  .next  
  app  
  (admin)  
    admin-dashboard  
    layout.tsx  
  (root)  
  api  
  admin  
  appointments  
  booking  
  route.ts  
  assessments  
  analytics  
  submit  
  templates  
  route.ts  
  auth  
  counselors  
  dashboard \ stats  
  route.ts  
  resources  
  route.ts  
  users \ counselors  
OUTLINE  
TIMELINE  
main  
app > (admin) > layout.tsx > ...  
26 import { Bell, LogOut, ChevronDown, Calendar, Shield } from "lucide-react";  
27 import { Button } from "@components/ui/button";  
28 import { useRouter } from "next/navigation";  
29 import { toast } from "sonner";  
30 import Link from "next/link";  
31 import { AppSidebar } from "@constants";  
32  
33 export default function DashboardLayout({ children }: { children: ReactNode }) {  
34   const [user, setUser] = useState<any>(null);  
35   const [isLoading, setIsLoading] = useState(true);  
36   const router = useRouter();  
37  
38   useEffect(() => {  
39     const checkUser = async () => {  
40       try {  
41         const res = await fetch("/api/auth/profile");  
42         const data = await res.json();  
43         console.log("Fetched user data:", data);  
44         if (data.user) {  
45           setUser(data.user);  
46         } else {  
47           router.push("/sign-in");  
48         }  
49       } catch (error) {  
50         console.error("Auth check failed:", error);  
51       }  
52     }  
53   });  
54 }  
DABIE GORDEN (4 days ago) Ln 30, Col 30 Spaces: 2 UTF-8 CRLF {} TypeScript JSX Go Live Prettier  
81° ENG INTL 2:35 PM 1/9/2026
```

Figure 5.4: Sample source code for the notification.js – Notification Page